

Gauge Carriers: Photon, Gluon, and Weak-Boson Standing

Field Notation, Gauge Redescription, and No Primitive Gauge-Boson Skin Escape in the Standard Model

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Abstract

This paper is SM-2 in the Standard Model carrier-exhaustion sequence. It imports the SM-1 master classifier and applies it to the gauge sector. The target is not to replace gauge theory, compute amplitudes, derive gauge-coupling values, or deny standard photon, gluon, W, and Z phenomenology. The target is local primitive-status closure for gauge standing. Gauge fields, gauge choices, potentials, propagators, polarizations, pole masses, widths, jets, detector events, color-flow displays, longitudinal/Goldstone descriptions, and gauge-sector EFT coefficients do not acquire primitive gauge-boson standing by themselves. Their standing is licensed through the closed gauge carrier network: gauge-order standing, standing-preserving transport, curvature/holonomy obstruction, charge-response standing, non-Abelian color-transport standing, weak-chiral response standing, anomaly-obstruction standing, and gauge-vacuum compatibility. The proof stack consists of the SM-1 classifier, R8, the A1–A6 Gauge Realization ladder, the PSM1–PSM6 bridge arc, and the SR1–SR7 Standard-Model Realization Rigidity arc. The color and electroweak certificates additionally use the Yang–Mills endpoint package, the L1 electroweak/equivalence package, and Higgs/mass-fixation support. From these inputs the paper proves that the gauge carrier certificates close and therefore that no residual same-scope primitive gauge-sector attribute survives outside the registered gauge carrier network. This is a scoped gauge-sector primitive-status result. It is not numerical parameter closure, not a model-by-model BSM catalogue, and not the global SM-6 endpoint.

The AASC/kernel dependency used here is not an interpretive overlay on gauge physics. It is the minimal non-degeneracy condition under which gauge-sector claims preserve standing, reference, sameness, comparison, prediction, and falsification. Gauge closure is therefore a same-scope primitive-status exhaustion theorem for gauge standing, not a vocabulary preference imposed on ordinary gauge practice.

Keywords: AASC; Standard Model; gauge carriers; photon; gluon; W boson; Z boson; gauge redescription; charge response; color transport; anomaly obstruction; gauge-vacuum compatibility; primitive closure; derivative skin; BSM audit.

Reading roadmap

SM-2 proceeds in five stages. Sections 1–3 fix the claim boundary, source dependencies, and gauge-sector classifier grammar. Sections 4–11 use the R8, A1–A6, PSM1–PSM6, SR1–SR7, Yang–Mills endpoint,

L1 equivalence, and Higgs/mass-fixation payloads that support scoped closure. Sections 12–14 classify photon, gluon, W, and Z standing. Sections 15–20 handle gauge couplings as an SM-4 boundary result, BSM/EFT branch certificates, the prediction box, scoped local closure, and the hostile reconstruction. The appendices provide the dependency ledger, carrier register, skin atlas, certificates, theorem inventory, and SM-6 handoff package.

1 Reader Orientation: What SM-2 Proves

1.1 One-sentence thesis

Gauge bosons are not primitive because they are field quanta; their standing is carrier-indexed through gauge-order, charge-response, color-transport, and weak-chiral roles.

This sentence is not a rejection of gauge theory. It is a primitive-status claim. The familiar field-theoretic descriptions remain valid as licensed representatives, measurement witnesses, transport ledgers, route statuses, and projection displays. What is denied is the automatic promotion of a gauge-sector skin into primitive standing.

1.2 SM-2 in the six-paper sequence

SM-1 built the master classifier, no-promotion discipline, target-update criterion, burden rule, and prediction register. SM-2 is the first sector-closure paper. It applies that machinery to gauge claims and returns a local carrier package to SM-6. The sector package is

$$C_{\text{gauge}} = \{C_{\text{gauge-order}}, C_{\text{transport}}, C_{\text{curv/hol}}, C_{\text{charge-response}}, \\ C_{\text{color-transport}}, C_{\text{weak-chiral}}, C_{\text{anomaly}}, C_{\text{gauge-vacuum}}\}.$$

SM-2 does not attempt the final SM-wide theorem

$$\text{Survprim}_{D_{\text{SM}}}(\text{Attr}_{\text{SM}} \setminus C_{\text{SM}}) = 0.$$

That is the SM-6 synthesis target after gauge, matter, charge/coupling/RG, flavor/mixing/generation, Higgs, BSM/EFT, and projection-interface closures are assembled.

1.3 Standard usage preserved

Standard physics language is preserved. The paper does not deny photons, gluons, W bosons, Z bosons, fields, curvature, polarizations, propagators, cross sections, decay widths, jets, detector events, or gauge-sector EFT methods. It denies only the primitive-status inference

$$\text{field/label/measurement/route or projection} \Rightarrow \text{primitive gauge carrier}.$$

1.4 Nonclaims

SM-2 does not claim:

- that gauge symmetry is ontic;
- that gauge potentials, gauges, or field representatives are primitive objects;
- that numerical gauge couplings g_1, g_2, g_3, e, α are derived;
- that gauge-sector BSM model-building is useless;
- that all named BSM models are catalogued or phenomenologically adjudicated;
- that the global SM primitive-exhaustion endpoint is already closed.

SM-2 claims instead that the local gauge-sector primitive-status question closes under the declared same-scope gauge arena and the carrier certificates listed below: certified primitive gauge standing descends to C_{gauge} , while field notation, gauge choice, measurement route, projection display, coupling value, and model label remain licensed derivative statuses unless a same-scope hostile carrier certificate closes outside C_{gauge} .

2 Kernel Status: Non-Degeneracy, Not Overlay

Gauge closure is not obtained by imposing an external vocabulary on gauge theory. The AASC/kernel supplies the minimal conditions under which a gauge claim can remain the same gauge claim through gauge choice, field notation, transport representation, curvature display, detector route, EFT projection, or BSM proposal. Without preserved target, standing-bearing reference, admissible redescription, and comparison class, the words “same photon claim”, “same gluon claim”, “same weak-boson claim”, or “same gauge-sector counterexample” lose determinate content.

Definition 2.1 (Non-degenerate gauge claim). A gauge-sector claim is non-degenerate only when its target, reference-bearing status, admissible comparison class, and standing conditions are preserved. In notation,

$$\text{NonDegenerateClaim}_{\text{gauge}}(x) \Rightarrow \text{KernelPreserved}_{\text{gauge}}(x).$$

The implication is not a convention of this paper; it records the necessity condition for the claim to remain an identifiable object of assertion, prediction, comparison, or failure.

Theorem 2.2 (Sameness requires kernel preservation). *For SM-2 claims in D_{gauge} , sameness is kernel-preserved sameness:*

$$\text{SameClaim}_{\text{gauge}}(x, y) \Rightarrow \text{KernelPreserved}_{\text{gauge}}(x, y).$$

Consequently,

$$\neg \text{KernelPreserved}_{\text{gauge}}(x, y) \Rightarrow \neg \text{SameClaim}_{\text{gauge}}(x, y).$$

A non-preserved target, comparison class, admissibility rule, or reference-bearing status yields a replacement, degeneration, target update, or failed comparison rather than the same gauge claim under test.

Theorem 2.3 (Prediction and falsification require non-degenerate reference). *Every prediction or falsification claim in the gauge arena presupposes preserved target identity, comparison class, and reference standing:*

$$\begin{aligned} & \text{Prediction}_{\text{gauge}}(x) \vee \text{Falsification}_{\text{gauge}}(x) \\ \Rightarrow & \text{KernelPreserved}_{\text{gauge}}(x) \wedge \text{TargetPreserved}(x) \wedge \text{ComparisonClassPreserved}(x). \end{aligned}$$

Gauge prediction and falsification require target-preserving comparison. Detector events, polarization data, jet structures, gauge-fixing displays, and EFT deviations may be physically decisive, but their evidential force is measured against a preserved gauge target rather than by automatic primitive promotion.

Proposition 2.4 (Kernel-denial burden). *A denial of the kernel is not a successful rejection of SM-2 unless it recovers the functions that the kernel supplies. Formally, any denial must satisfy the burden*

$$\begin{aligned} \text{KernelDenied}(x) \Rightarrow & \text{RecoverSameness}(x) \wedge \text{RecoverStanding}(x) \\ & \wedge \text{RecoverReference}(x) \wedge \text{RecoverFalsification}(x). \end{aligned}$$

Thus a gauge counterexample must preserve the same target, same comparison class, and same reference-bearing object while showing that gauge-order, transport, curvature/holonomy, charge response, color

transport, weak-chiral response, anomaly-compatibility, and gauge-vacuum standing can be given without the kernel conditions used here.

Remark 2.5 (Proof class). The result is not a Lagrangian reconstruction, not a numerical parameter derivation, not an empirical-finality theorem, and not a no-new-physics theorem. It is a same-scope primitive-status exhaustion theorem for the stated gauge arena. New measurements, anomalies, effective descriptions, and target-changing discoveries remain admissible; they do not become unlicensed same-scope primitive gauge carriers merely by being real or useful.

3 Dependency Ledger and Proof-Class Boundary

3.1 Upstream kernel dependency and source direction

The dependency order is fixed as follows. The kernel/admissibility layer supplies the minimal non-degeneracy conditions for sameness, standing, reference, comparison, prediction, and falsification. SM-2 instantiates those conditions in the gauge arena; it does not derive the kernel from the Standard Model or from any later structural-spine paper. Later Standard Model structural results may instantiate the same kernel at larger scope, but they do not supply the upstream necessity of kernel preservation.

Accordingly, dependence on the kernel is not a defect in the proof class. A successful objection must either break a received local closure premise, supply a faithful same-scope primitive certificate, or recover non-degenerate physical claimhood without the kernel functions listed in Proposition 2.

3.2 Imported SM-1 machinery

SM-2 receives the SM-1 classifier normal form

$$R_x = (D_{\text{SM}}, x, C_x, \text{Role}_x, \text{Rep}_x, \text{Meas}_x, \text{Transport}_x, \text{Route}_x, \text{Mix}_x, \text{Proj}_x, \\ \text{Extension}_x, \text{Update}_x, \text{Cert}_x, \rho_x, \text{Pred}_x).$$

The imported SM-1 results are used as classifier machinery, not rederived:

- carrier-admission conditions and carrier-certificate sufficiency;
- non-circular skin-identification tests;
- no primitive skin promotion when skin criteria hold and no carrier certificate closes;
- target-update discipline;
- counterexample burden as a review state, not a survivor state;
- non-ad-hoc classifiability, with failure of P0 as the exposed failure path;
- the prediction-register format and falsification protocol.

3.3 Gauge dependency package

The local proof uses the following stable gauge-sector source layers.

Layer	Source role	Use in SM-2
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R8	Gauge structure from boundary-trace fixation	Grounds gauge structure as boundary-trace-fixed presentation equivalence rather than primitive symmetry.
A1	Gauge Transition Groupoid	Establishes boundary-fixed local/restriction transition organization.
A2	Connection as Standing-Preserving Transport Geometry	Establishes transport-comparison role.
A3	Curvature as Transport-Holonomy Obstruction	Establishes boundary-detectable loop residue and obstruction standing.
A4	Representation Admissibility and Charge-Witness Rigidity	Establishes representation-response and charge-witness roles.
A5	Anomaly as Standing-Failure Obstruction	Establishes anomaly cancellation as admissibility requirement for representation-bearing gauge standing.
A6	Gauge-Vacuum Compatibility	Establishes gauge-vacuum compatibility and symmetry-breaking admissibility as response-reduction discipline.
SR1–SR7	Standard-Model Realization Rigidity arc	Supplies simultaneous constraint collapse, product, representation, charge, anomaly, residual/vacuum compression, and final SM survivor assembly.
Gauge ontology	Why Gauge Symmetry Cannot Be Ontic	Blocks primitive-ontology reading of gauge symmetry.
Coupling status	Gauge-Coupling Parameters and Dimensionless Couplings as Moduli	Blocks primitive promotion of coupling values and RG coordinates.
PSM1–PSM6	Lie-type, product, representation, Abelian charge, anomaly, and vacuum-compatible residual bridge arc	Fills the bridge from A1–A6 gauge realization to SR1–SR7 Standard-Model survivor rigidity.
Yang–Mills endpoint package	Vacuum-sector mass gap, proof kernel, Companions I–III, and endpoint well-definedness	Discharges the color-transport/gluon carrier burden without making SM-2 a QCD monograph.
L1 equivalence package	Charged-current collapse, EW/EM bookkeeping, longitudinal/Goldstone equivalence, hypercharge/electric charge, anomaly/Yukawa, FCNC, equivalence-space closure	Discharges the weak-boson/electroweak, charge-response, and longitudinal/Goldstone non-promotion burdens.
Higgs/mass fixation	Higgs bridge/fixation arc and mass-fixation support	Licenses weak-boson mass-response standing while keeping vev, pole masses, and widths derivative.

3.4 Theorem-status ladder

The paper uses the following status labels.

Status	Meaning in SM-2
Definition	A stipulated object in the gauge-sector classifier grammar.
Criterion	A rule for admitting or rejecting a gauge-sector status assignment.
Imported theorem	A result used from SM-1, R8, A1–A6, PSM1–PSM6, SR1–SR7, the Yang–Mills endpoint package, the L1 equivalence package, or the Higgs bridge/fixation arc.
Closed SM-2 theorem	A local gauge-sector result proved by applying the imported machinery to D_{gauge} .
Sector-local corollary	A gauge result handed forward to SM-3, SM-4, SM-5, or SM-6.
Open burden	A live gauge-sector challenge lacking a closed carrier certificate or target-update exclusion.
SM-6 endpoint target	A global primitive-status deletion theorem not claimed in SM-2.

Throughout SM-2, *closed* means closed for primitive-status classification in the declared same-scope gauge arena. It does not mean numerical phenomenological closure, empirical finality, or model-by-model BSM classification.

3.5 Central theorem family

The local theorem family is:

1. Gauge Carrier Normal Form Theorem.
2. Gauge Skin Non-Promotion Theorem.
3. Boundary-Trace Gauge Role Theorem.
4. Gauge Transition Carrier Theorem.
5. Connection Transport Carrier Theorem.
6. Curvature/Holonomy Obstruction Carrier Theorem.
7. Representation/Charge Carrier Theorem.
8. Anomaly Obstruction Carrier Theorem.
9. Gauge-Vacuum Compatibility Carrier Theorem.
10. Photon Charge-Response Carrier Theorem.
11. Gluon Non-Asymptotic Color-Transport Carrier Theorem.
12. Weak-Boson Weak-Chiral Carrier Theorem.
13. Gauge-Coupling Non-Carrier Proposition.
14. Gauge BSM Branch-Certificate Theorem.
15. Gauge Prediction Box Theorem.

3.6 Source-status convention for SM-2 theorems

Every major theorem below carries a source-status tag. When a theorem uses R8, A1–A6, PSM1–PSM6, SR1–SR7, the Yang–Mills endpoint package, the L1 equivalence package, the Higgs bridge/fixation arc, the mass-fixation support, or the SM-1 classifier, SM-2 does not reprove that source theorem. SM-2’s contribution is the gauge-sector carrier/skin classification theorem produced by applying the imported payload to D_{gauge} . The closure claim is that the relevant gauge carrier certificates close under the received dependency package.

Theorem layer	Source status	SM-2 contribution
Boundary trace	Imported from R8	Gauge-order carrier classification.
Transition groupoid	Imported from A1	Gauge presentation as licensed skin, not primitive occupant.
Connection transport	Imported from A2	Gauge potentials as transport representatives.
Curvature/holonomy	Imported from A3	Field strength / loop residue as obstruction display.
Rep./charge	Imported from A4, PSM4, SR3/SR4, and L1 charge support	Charge-response and photon carrier closure.
Anomaly	Imported from A5, PSM5, SR5, and L1 anomaly/Yukawa support	Anomaly ledger as standing-failure obstruction.
Gauge-vacuum	Imported from A6, PSM6, SR6	Electroweak split and gauge-vacuum carrier boundary.
PSM bridge	Imported from PSM1–PSM6	Lie, product, representation, charge, anomaly, and vacuum normal-form bridge from A-ladder to SR survivor results.
Gauge-product survivor	Imported from SR1–SR7	Recovered SM gauge product as survivor presentation.
Yang-Mills color	Imported from YM endpoint package	Color-transport/gluon carrier closure and non-asymptotic primitive-status discipline.
L1 EW equivalence	Imported from L1 package	Charged-current, EW/EM, longitudinal and Goldstone, charge, anomaly/Yukawa, and equivalence-space closure.
Photon, gluon, W/Z	Local SM-2 closure certificates	Gauge-boson primitive-status closure.

4 Gauge Closure Proof Spine

4.1 Stable closure route

The gauge-sector primitive-status closure proof spine is displayed before the component sections so that the reader can audit the scope of the final theorem. The component certificates are discharged in

Sections 12–17 and summarized again in the final closure section. The route is:

$$SM1 + R8 + A_{1-6} + PSM_{1-6} + SR_{1-7} + YM_{\text{endpoint}} + L1_{\text{EW}} + H_{\text{br/fix}} \Rightarrow \text{ClosedGCert}(C_{\text{gauge}}).$$

Here YM_{endpoint} abbreviates the Yang–Mills color endpoint package: the vacuum-sector mass-gap paper, proof kernel, companions I–III, and endpoint well-definedness paper. The symbol $L1_{\text{EW}}$ abbreviates the L1 equivalence package: charged-current collapse, EW/EM bookkeeping, longitudinal/Goldstone equivalence, hypercharge/electric charge quantization, anomaly freedom versus Yukawa realizability, neutral-current/FCNC discipline, mass-fixation support, and equivalence-space closure.

4.2 Closure spine proposition

Proposition 4.1 (Gauge Closure Proof-Spine Proposition).

$$SM1 + R8 + A_{1-6} + PSM_{1-6} + SR_{1-7} + YM_{\text{endpoint}} + L1_{\text{EW}} + H_{\text{br/fix}} \Rightarrow \text{ClosedGCert}(C_{\text{gauge}}).$$

The gauge carrier package

$$C_{\text{gauge}} = \{C_{\text{gauge-order}}, C_{\text{transport}}, C_{\text{curv/hol}}, C_{\text{charge-response}}, \\ C_{\text{color-transport}}, C_{\text{weak-chiral}}, C_{\text{anomaly}}, C_{\text{gauge-vacuum}}\}$$

is the closed gauge carrier package. The proposition states the proof architecture. The final carrier-certificate closure theorem is restated after the photon, gluon, weak-boson, coupling-boundary, and BSM branch sections.

4.3 Certificate-to-primitive-status deletion form

Lemma 4.2 (Certificate-to-Primitive-Status Deletion Form).

$$\text{ClosedGCert}(C_{\text{gauge}}) \Rightarrow \text{Survprim}_{D_{\text{gauge}}}(\text{Attr}_G \setminus C_{\text{gauge}}) = 0.$$

Once the gauge carrier certificates close, a same-scope gauge-sector claim outside C_{gauge} has only registered statuses left: licensed representative, measurement witness, transport ledger, route status, projection display, extension disposition, target update, counterexample burden, or failed primitive. A closed primitive survivor outside C_{gauge} is exactly the hostile falsifier described later.

5 The Gauge-Sector Claim Arena

Definition 5.1 (Gauge-sector arena). The gauge-sector claim arena is the gauge subarena of the fixed Standard Model comparison domain:

$$D_{\text{gauge}} = (D_{\text{SM}}, G_{\text{SM}}, R_{\text{SM}}, Q_{\text{SM}}, A_{\text{anom}}, V_{\text{SM}}, H_{\text{br/fix}}, \text{Transport}_{\partial D}, \text{GaugeAdm}).$$

It includes the recovered gauge-product role, representation-response skeleton, charge lattice, anomaly ledger, vacuum-compatible residual form, Higgs bridge/fixation boundary for weak-boson standing, and gauge transport/boundary-trace discipline.

Definition 5.2 (Gauge claim). $\text{GaugeClaim}(x)$ means that x is a submitted claim about a gauge field, gauge potential, gauge boson, photon, gluon, W/Z boson, gauge group, gauge choice, polarization, propagator, ghost/gauge-fixing artifact, charge/current, coupling value, RG flow, anomaly, vacuum-breaking route, or gauge-sector BSM/EFT extension.

Definition 5.3 (Same-scope gauge claim). $\text{SameScope}_{D_{\text{gauge}}}(x)$ holds when x preserves D_{SM} , the recovered gauge-product role, representation-response skeleton, charge lattice, anomaly ledger, vacuum compatibility, Higgs bridge/fixation boundary when relevant, and the no-hidden-load condition. A claim that changes one of those items is evaluated as target update rather than same-scope gauge survivor.

Criterion 5.4 (Gauge target update). A gauge claim is a target update only if it changes gauge-product carrier inventory, representation ledger, charge lattice, anomaly ledger, vacuum regime, Higgs/fixation boundary, source state, comparison arena, hidden-sector carrier inventory, or the regime conditions under which the claim is evaluated.

6 Gauge Carrier Admission and Gauge Skin Identification

Definition 6.1 (Gauge carrier certificate). A closed gauge carrier certificate is not a synonym for carrier status. It is the package of local obligations whose closure suffices for carrier status:

$$\begin{aligned} \text{GCert}(x) &:= \text{RoleCl}(x) \wedge \text{RedInv}(x) \\ &\quad \wedge \text{Lic}(x) \wedge \text{NoDesc}(x) \\ &\quad \wedge \text{NoSel}(x) \wedge \text{GCclose}(x). \end{aligned}$$

Here RoleCl closes the load-bearing gauge role, RedInv proves invariance under lawful gauge/basis/scheme/projection changes, Lic proves that the familiar descriptions are licensed by the role, NoDesc excludes derivative descent when primitive status is asserted, NoSel blocks a derivative selector from entering as primitive, and GCclose supplies the relevant local gauge-sector theorem.

Proposition 6.2 (Certificate sufficiency).

$$\text{GCert}(x) \Rightarrow \text{Carrier}_{\text{gauge}}(x).$$

The implication is non-circular because the certificate does not include $\text{Carrier}_{\text{gauge}}(x)$ as a premise.

Lemma 6.3 (Gauge skin identification). *A submitted gauge-sector item s is eligible for gauge-skin classification when its status depends on at least one of the following: gauge choice, representation chart, basis, normalization, renormalization scheme, measurement route, detector route, perturbative expansion, projection interface, field label, model name, coupling coordinate, or failure of invariant role preservation under admissible redescription.*

Theorem 6.4 (Gauge Skin Non-Promotion).

$$\text{GaugeSkin}(s) \wedge \neg \text{GCert}(s) \Rightarrow \neg \text{Primitive}_{\text{gauge}}(s).$$

Thus a gauge potential, gauge fixing, polarization vector, propagator, pole mass, decay width, jet, detector event, color-flow diagram, coupling value, RG crossing, EFT coefficient, or model label cannot supply primitive gauge standing merely by being available in a valid calculation or measurement route.

Remark 6.5 (Local application). The lemma supplies the SM-2 eligibility test. Particular applications are discharged locally in this paper when the gauge dependency package closes the relevant role. Items whose status depends on charge/coupling/RG detail are handed to SM-4 after SM-2 blocks the primitive-gauge-carrier promotion.

Gauge carrier component	Closure support	Certificate output	Status
$C_{\text{gauge-order}}$	R8 + A1	boundary-fixed gauge role and local transition groupoid; no individual presentation is primitive.	closed
$C_{\text{transport}}$	A2	standing-preserving connection/transport comparison; potentials and propagators are representatives.	closed
$C_{\text{curv/hol}}$	A3	curvature/holonomy obstruction carrier; field-strength and loop displays are licensed.	closed
$C_{\text{charge-response}}$	A4 + PSM4 + SR4 + L1 hypercharge/electric charge + EW/EM bookkeeping	unbroken and representation-indexed charge-response standing; photon certificate closes in Section 12.	closed
$C_{\text{color-transport}}$	A4 + PSM/SR product and representation support + Yang–Mills endpoint package + proof kernel + Companions I–III + endpoint well-definedness	non-asymptotic color-transport standing; perturbative lines, color-flow diagrams, and jets are route or projection statuses.	closed
$C_{\text{weak-chiral}}$	A6 + PSM6 + SR6/SR7 + L1 charged-current collapse + longitudinal/Goldstone equivalence + EW/EM bookkeeping + Higgs/mass fixation	charged and neutral weak response standing after electroweak split, licensed by gauge-vacuum and Higgs bridge discipline.	closed
C_{anomaly}	A5 + PSM5 + SR5 + L1 anomaly freedom $\Leftrightarrow$ Yukawa realizability	anomaly as standing-failure obstruction and anomaly-neutral survivor ledger.	closed
$C_{\text{gauge-vacuum}}$	A6 + PSM6 + SR6 + Higgs bridge/fixation + mass fixation	vacuum-compatible gauge standing and broken/unbroken route discipline; vev-like quantities remain licensed statuses.	closed

7 Gauge Skin Atlas

The closure spine above fixes the carrier side of the argument. The following atlas records the main gauge-sector descriptions that remain valid but derivative. Each entry is licensed by the closed carrier network and therefore cannot self-promote into primitive gauge standing without an independent carrier

certificate.

Gauge-sector item	Expected status	Reason
Gauge potential A_μ	licensed representative / transport coordinate	depends on gauge presentation and connection chart.
Field-strength notation	curvature representative	displays loop-residue role; notation alone is not primitive.
Gauge fixing	skin / calculation route	selects representative without changing carrier standing.
Ghost fields	gauge-fixing calculational skin	live in gauge-fixed quantization route, not primitive gauge standing.
Propagator	route/calculation skin	depends on representation, gauge, scheme, and perturbative setup.
Polarization vector	representative / measurement skin	route-dependent display of physical response.
Pole mass	measurement witness / mass-response skin	witnesses weak-boson response; does not ground identity.
Decay width	route/measurement witness	downstream route property.
Jet	detector / projection route	evidences color transport but is not isolated gluon standing.
Color-flow diagram	projection/route display	useful bookkeeping over color transport.
Coupling value	transport/calibration ledger	not primitive gauge carrier; SM-4 expands this.
RG crossing	transport or projection display	does not generate primitive gauge standing.

8 Gauge Structure from Boundary Trace

8.1 R8 payload

R8 derives the minimal gauge role from boundary-trace-fixed presentation equivalence. Gauge structure is not first introduced as a primitive symmetry group. It is the role required when local physical presentations must preserve field-carrier mediation, particle-persistence identity, quotient comparison, bounded transport, and boundary witnesses under admissible redescription.

Theorem 8.1 (Boundary-Trace Gauge Role).

$$I_{\text{phys}}(D) \wedge \text{BTrace}(D) \Rightarrow C_{\text{gauge-order}}(D).$$

Status. Imported theorem from R8, specialized to D_{gauge} ; SM-2 uses it as the gauge-order carrier input rather than reproving the physical-realization bridge. This imported theorem establishes the gauge-order carrier role before the richer groupoid, connection, curvature, representation, anomaly, or vacuum layers are introduced.

Status: imported theorem from R8, specialized to D_{gauge} ; SM-2 uses it as the gauge-order carrier input.

Nonclaim 8.2 (Boundary-trace theorem boundary). The boundary-trace result does not derive a specific Lie group, the Standard Model gauge product, numerical gauge couplings, Yang-Mills dynamics, or individual photon/gluon/W/Z standing. It establishes the gauge-order carrier role to which later source layers attach.

9 Transition Groupoid and Gauge-Order Carrier

9.1 A1 payload

A1 proves that admissible local gauge transitions minimally admit a boundary-fixed local/restriction groupoid organization. Local gauge presentations cannot remain arbitrary pairwise identifications. Identity, inverse, restriction, partial composition, overlap coherence, and associativity are forced by role-preservation and admissible continuation.

Definition 9.1 (Transition carrier).

$$\mathbf{Gpd}_{\partial}(D)$$

denotes the boundary-fixed local/restriction transition organization for gauge presentation freedom.

Theorem 9.2 (Gauge Transition Carrier).

$$C_{\text{gauge-order}}(D) \Rightarrow \mathbf{Gpd}_{\partial}(D).$$

Status. Imported theorem from A1, applied as a closed SM-2 carrier-certificate component. The local transition carrier licenses local gauges, transition functions, patch data, and chart changes as valid representatives while preventing any one presentation from becoming the primitive gauge occupant.

Status: imported theorem from A1, specialized to D_{gauge} ; SM-2 derives the gauge-order skin-deletion certificate from it.

Certificate field	Content
Carrier	$C_{\text{gauge-order}}$ through $\mathbf{Gpd}_{\partial}(D)$.
Licensed skins	local gauges, charts, transition functions, generator notation.
False primitive deleted	gauge presentation as primitive physical occupant.
Falsifier	same-scope gauge standing outside boundary-fixed transition organization with closed carrier certificate.

10 Connection as Standing-Preserving Transport

10.1 A2 payload

A2 derives the standing-preserving transport-comparison role. The connection is not introduced as primitive gauge substance. It is the role required to compare gauge-presentation-dependent carrier content along admissible continuation morphisms without importing a hidden global selector.

Definition 10.1 (Connection transport role).

$$\nabla_{\partial D}$$

denotes the boundary-fixed standing-preserving comparison assignment along typed continuation morphisms.

Theorem 10.2 (Connection Transport Carrier).

$$\text{Gpd}_{\partial}(D) \Rightarrow \nabla_{\partial D}.$$

Status. Imported theorem from A2, specialized to D_{gauge} ; SM-2 contributes the no-promotion corollary for potentials, propagator setups, and transport representatives. The theorem licenses connection charts, potentials, covariant-derivative notation, and transport representatives as standing-preserving comparison data. It does not license gauge potential notation as primitive physical ontology.

Status: imported theorem from A2, specialized to D_{gauge} ; SM-2 contribution is the transport-skin non-promotion corollary.

Proposition 10.3 (Gauge-potential non-primitivity).

$$A_{\mu} \text{ notation} \wedge \neg \text{GCert}(A_{\mu}) \Rightarrow \neg \text{Primitive}_{\text{gauge}}(A_{\mu}).$$

The potential is a licensed representative of transport-comparison structure; it is not primitive gauge standing by notation alone.

11 Curvature and Holonomy as Obstruction Carrier

11.1 A3 payload

A3 derives the minimal loop-residue role later realized by curvature and holonomy structures. Nontrivial transport residue around loops has standing only when it is boundary-detectable, quotient-stable, same-domain, transport-witnessed, gauge-compatible, and standing-bearing.

Definition 11.1 (Curvature/holonomy obstruction role).

$$\Omega_{\partial D}$$

denotes the boundary-fixed assignment of admissible loop residues to endpoint-closed continuation paths.

Theorem 11.2 (Curvature/Holonomy Obstruction Carrier).

$$\nabla_{\partial D} \Rightarrow \Omega_{\partial D}.$$

Status. Imported theorem from A3, used as the closed obstruction-carrier component in SM-2. The carrier licenses field-strength notation, Wilson-loop-style displays, local curvature expressions, and obstruction calculations. The notation is not primitive by itself; the carrier is the boundary-detectable transport-holonomy residue.

Status: imported theorem from A3, specialized to D_{gauge} ; SM-2 contribution is the curvature/holonomy representative-status classification.

Certificate field	Content
Carrier	curvature/holonomy obstruction carrier.

Licensed skins	$F_{\mu\nu}$, local curvature notation, Wilson-loop-style displays, field-strength charts.
False primitive deleted	field-strength formula as primitive object.
Falsifier	same-scope obstruction standing outside transport/holonomy residue with closed carrier certificate.

12 Representation and Charge-Witness Rigidity

12.1 A4 and SR3/SR4 payload

A4 derives representation-response and charge-witness structure as standing-preserving response roles. SR3 and SR4 later sharpen this into representation-response and Abelian charge-ratio rigidity in the admissible gauge arena. Representation labels, charge numbers, bases, dimensions, normalizations, and raw internal indices are not primitive physical structure. They must descend to witness-bearing response roles.

Definition 12.1 (Representation and charge-witness roles).

$$\text{Rep}_{\partial D}, \quad \text{ChWit}_{\partial D}$$

denote the representation-response role and boundary-detectable charge-witness role.

Theorem 12.2 (Representation/Charge Carrier).

$$\Omega_{\partial D} \Rightarrow (\text{Rep}_{\partial D}, \text{ChWit}_{\partial D}).$$

Status. Imported theorem from A4, sharpened by SR3/SR4. SM-2 uses it as a closed carrier-certificate component for photon charge-response and as a handoff boundary for matter-response papers. Representation-response and charge-witness standing enter as quotient-stable response structure, not arbitrary labels, hidden multiplicities, basis choices, or normalization conventions.

Status: imported theorem from A4 with SR3/SR4 rigidity support, specialized to D_{gauge} ; SM-2 uses it for photon charge-response and representation-skin classifications.

Proposition 12.3 (Charge number non-primitivity).

$$\text{ChargeNumber}(q) \wedge \neg \text{GCert}(q) \Rightarrow \neg \text{Primitive}_{\text{gauge}}(q).$$

A charge number can witness response; it cannot be primitive charge-response standing by numerical label alone.

13 Anomaly as Standing-Failure Obstruction

13.1 A5 and SR5 payload

A5 proves that anomaly-obstruction cancellation is an admissibility requirement for representation-bearing gauge realization. SR5 then collapses anomaly ledgers over the charge survivor arena. An anomaly is not merely an algebraic nuisance; it marks local-to-global standing failure when gauge-related descriptions assemble into different boundary-detectable ledgers.

Definition 13.1 (Anomaly-obstruction role).

$$\text{Anom}_{\partial D}$$

denotes the standing-obstruction role later realized by local, global, mixed, current, phase, measure, and cohomological anomaly formalisms.

Theorem 13.2 (Anomaly Obstruction Carrier).

$$(\text{Rep}_{\partial D}, \text{ChWit}_{\partial D}) \Rightarrow \text{Anom}_{\partial D}, \quad \text{Anom}_{\partial D} = 0 \text{ for survivor gauge standing.}$$

Status. Imported theorem from A5, sharpened by SR5. SM-2 uses it to close the anomaly-obstruction certificate and to discipline extra gauge-factor branches. The theorem classifies anomaly-cancellation conditions as downstream specializations of obstruction-vanishing, not as primitive algebraic objects.

Status: imported theorem from A5 with SR5 rigidity support, specialized to D_{gauge} ; SM-2 uses it as the anomaly-obstruction ledger.

Proposition 13.3 (Anomalous same-scope failure). A same-scope representation-bearing gauge claim with nontrivial anomaly obstruction fails gauge standing unless it supplies a target update, an anomaly-resolution certificate under a changed arena, or a closed same-scope carrier certificate defeating the obstruction role.

14 Gauge-Vacuum Compatibility

14.1 A6 and SR6 payload

A6 proves that gauge-vacuum compatibility is an admissibility requirement for anomaly-neutral representation-bearing gauge structures. SR6 later collapses residual/vacuum-sector normal forms. The symmetry-breaking admissibility role is vacuum-mediated response reduction. It is not primitive gauge destruction, arbitrary vacuum selection, or global representative choice.

Definition 14.1 (Gauge-vacuum compatibility).

$$\text{GaugeVac}_D$$

denotes the vacuum-compatible residual gauge-standing role preserving transition organization, transport comparison, loop residue, representation response, charge witness, anomaly neutrality, field-carrier mediation, particle persistence, quotient comparison, bounded transport, boundary trace, deformation baseline, and restoration baseline.

Theorem 14.2 (Gauge-Vacuum Compatibility Carrier).

$$\text{Anom}_{\partial D} = 0 \Rightarrow \text{GaugeVac}_D$$

inside the declared realized physical envelope. **Status.** Imported theorem from A6, sharpened by SR6, and used in SM-2 as the gauge-vacuum carrier-certificate component for weak-boson standing. The carrier licenses broken/unbroken phase descriptions, residual-sector displays, vacuum-mediated response-reduction language, and the bridge toward Higgs-like mass-response standing.

Status: imported theorem from A6 with SR6 rigidity support, specialized to D_{gauge} ; SM-2 uses it for electroweak split and weak-boson carrier certificates.

Nonclaim 14.3 (Vacuum compatibility boundary). This theorem does not make a vacuum coordinate, vev-like coordinate, order parameter, scalar offset, or residual subgroup notation primitive. Those are licensed statuses only when they descend to gauge-vacuum and Higgs bridge/fixation discipline.

15 Recovered Gauge Product and Gauge-Product Survivor

15.1 SR1–SR7 payload

The Standard-Model Realization Rigidity arc begins after the gauge-realization and PSM bridge layers have licensed a Standard-Model-facing gauge arena. SR1 proves simultaneous constraint collapse over product, representation, charge, anomaly, vacuum, residual, transport, and deformation ledgers. SR2 compresses product roles. SR3 compresses representation response. SR4 compresses Abelian charge roles. SR5 compresses anomaly ledgers. SR6 compresses residual/vacuum sectors. SR7 assembles the joint survivor.

The resulting chain is

$$\begin{aligned} \text{BTrace}(D) &\Rightarrow \text{Gpd}_\partial(D) \Rightarrow \nabla_{\partial D} \Rightarrow \Omega_{\partial D} \\ &\Rightarrow (\text{Rep}, \text{ChWit}) \Rightarrow \text{Anom} = 0 \Rightarrow \text{GaugeVac}_D \\ &\Rightarrow G_{\text{SM}} / \simeq_{\text{Adm}} . \end{aligned}$$

Theorem 15.1 (Gauge-Product Survivor). *Within the SR7 comparison regime, the survivor profile of the gauge arena recovers the Standard Model gauge-product presentation*

$$G_{\text{SM}} = \text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y$$

up to admissible redescription and declared normalization conventions.

Status: *imported theorem from SR1–SR7, used as the gauge-product survivor payload; SM-2 does not rederive SR7.*

Nonclaim 15.2 (Gauge-product versus coupling values). The gauge-product survivor theorem does not derive the numerical values of g_1, g_2, g_3, e, α , running coupling values, or unification-scale relations. These belong to coupling/transport/moduli discipline and are treated fully in SM-4.

16 Photon Standing

16.1 Submitted photon claims

Photon-associated submissions include claims about $A_\mu, F_{\mu\nu}$, polarization, electromagnetic radiation, charge-current coupling, masslessness, Coulomb display, detector events, and photon field quanta. SM-2 classifies these claims by asking whether the submission preserves D_{gauge} , whether it descends to charge-response and unbroken electromagnetic transport, and whether it supplies any carrier certificate independent of those roles.

Definition 16.1 (Photon carrier).

$$C_\gamma = C_{\text{charge-response}} \wedge C_{\text{unbroken-EM transport}} .$$

The photon carrier is the unbroken electromagnetic charge-response transport role inside the recovered gauge arena. It is not the gauge potential, a polarization vector, a detector event, a coupling value, or a field-quantum label by itself.

16.2 Photon carrier certificate

Certificate field	Photon closure content
Fixed target	D_{gauge} with recovered gauge product, charge lattice, anomaly ledger, vacuum compatibility, and same-scope boundary discipline preserved.
Carrier role	unbroken electromagnetic charge-response transport: a standing-preserving response role licensed by representation/charge-witness rigidity and gauge-vacuum residual compatibility.
Source payload	R8 supplies gauge-order role; A1–A3 supply transition, transport, and curvature/holonomy carriers; A4 plus SR3/SR4 supply representation/charge-witness rigidity; A6/SR6 supply the residual unbroken route; SR7 supplies the gauge-product survivor.
Licensed skins	A_μ , $F_{\mu\nu}$, polarization, Coulomb display, radiation modes, charge-current descriptions, detector events, and low-energy electromagnetic projections.
No-hidden-selector proof	no gauge choice, polarization convention, coupling value, measurement route, or detector projection is permitted to select primitive photon identity.
Descent exclusion	a primitive photon claim that refuses descent to charge-response/unbroken-EM transport must supply a same-scope carrier certificate outside C_γ .
Falsifier	a same-scope photon primitive outside charge-response/gauge-order standing that preserves D_{gauge} and closes a carrier certificate.

Theorem 16.2 (Photon Charge-Response Carrier). *If $\text{PhotonClaim}(x) \wedge \text{SameScope}_{D_{\text{gauge}}}(x)$, then x is classified as a carrier only by descent to C_γ or by supplying a closed hostile carrier certificate outside C_γ . Otherwise*

$$\rho_{\text{gauge}}(x) \in \{\text{Rep, Meas, Trans, Route, Proj, Burden, FailPrim}\}.$$

Status: closed SM-2 theorem relative to the R8/A1–A4/A6/SR4/SR6/SR7 photon certificate above.

Photon-associated item	Status	Reason
Gauge potential A_μ	transport representative	depends on gauge/chart choice.
Field strength $F_{\mu\nu}$	curvature representative	displays transport-holonomy residue.
Polarization	representative / measurement skin	depends on mode, basis, and route.
Detector event	measurement witness	witnesses response; does not ground primitive identity.
Coulomb display	projection / route display	low-energy display of charge response.

Masslessness statement	structural/residual route status	descends to unbroken residual electromagnetic standing.
Charge-current interaction	charge-response witness	evidences C_γ , not a separate primitive.
Coupling e	transport/calibration ledger	SM-4 expands positive coupling theory.

Criterion 16.3 (Photon falsifier). A photon-sector falsifier must preserve D_{gauge} , refuse descent to unbroken electromagnetic charge-response transport, exclude gauge/measurement/projection/transport status, and close an independent carrier certificate outside C_γ .

17 Gluon Standing

17.1 Submitted gluon claims

Gluon-associated submissions include claims about gluon field notation, color labels, perturbative gluon lines, jets, color-flow diagrams, confinement routes, hadronization displays, non-Abelian charge flow, Wilson or loop displays, local gauge-invariant Yang–Mills constructions, and asymptotic free-gluon standing.

Definition 17.1 (Gluon carrier).

$$C_g = C_{\text{color-transport}}.$$

Gluon standing is non-Abelian color-transport standing. The primitive-status claim is not supplied by a perturbative line, a color index, a jet, a parton-shower display, or a color-flow diagram. Those are licensed representatives, route statuses, or projections of the color-transport carrier.

17.2 Gluon color-transport carrier closure from the Yang–Mills endpoint package

The Yang–Mills endpoint package supplies the color-sector outputs used by SM-2 to close the color-transport carrier certificate. SM-2 does not reproduce that proof package. It imports the endpoint outputs listed below to discharge the primitive-status burden for gluon standing.

Yang–Mills source	Imported payload used by SM-2	Source anchor
Vacuum-sector mass gap for the local gauge-invariant sharp-local Yang–Mills net	Vacuum-sector separation and nontrivial local gauge-invariant standing used to support color-transport carrier closure.	Main vacuum-sector closure theorem and endpoint certificate.
Proof Kernel for the Local Gauge-Invariant Sharp-Local Yang–Mills Vacuum-Sector Mass-Gap Theorem	Kernel certificate for the sharp-local vacuum-sector theorem.	Proof-kernel theorem and certificate ledger.
Companion I: Ultraviolet Gate and Route 1 Mass-Gap Chain	Ultraviolet gate and Route 1 mass-gap support chain.	Route 1 chain theorem.

Companion II: Lane A Sharp-Local Construction	Sharp-local construction support for the local gauge-invariant net.	Lane A construction theorem.
Companion III: Reconstruction, Non-Triviality, and Faithful-Wilson Universality	Reconstruction and faithful Wilson-universality support.	Reconstruction and faithful-Wilson universality theorem.
Admissibility and the Well-Definedness of Yang–Mills Endpoint Constructions	Determinacy, identity, and uniqueness discipline for Yang–Mills endpoint constructions.	Endpoint well-definedness theorem.

The imported payload is used narrowly. A perturbative gluon line remains a calculation route; a color-flow diagram remains a projection/bookkeeping display; a jet remains a detector and hadronization route; a Wilson or loop display remains a witnessed obstruction or route display. Their evidentiary value is preserved, but primitive isolated-gluon standing is not generated by any one of them. The carrier role closed for SM-2 is color transport, not free-particle skin.

Certificate field	Gluon certificate content
Fixed target	D_{gauge} , with the recovered non-Abelian color role, product-sector ledger, representation-response structure, anomaly ledger, and gauge-vacuum compatibility preserved.
Carrier role	Non-Abelian color-transport standing, supported by the Yang–Mills endpoint package and the A4/PSM/SR response payload.
Source payload	R8/A1–A4 for gauge-order, transport, curvature, representation, and response standing; PSM/SR for product and representation realization; Yang–Mills endpoint package for local gauge-invariant color-sector closure; QCD CP-odd equivalence as supplementary bookkeeping where relevant.
Licensed skins	Gluon field notation, color index labels, perturbative gluon lines, color-flow diagrams, parton-level route language, Wilson or loop displays, hadronization descriptions, and jet signatures.
No-hidden-selector proof	No perturbative gauge choice, line in a diagram, color label, parton shower display, Wilson display, or detector jet selects primitive isolated-gluon identity.
Descent exclusion	A primitive free-gluon claim must defeat descent to color-transport standing and must not be merely a route, projection, or target-updating regime change.
Falsifier	A free gluon with same-scope primitive SM standing outside $C_{\text{color-transport}}$, with target update and derivative descent excluded and a closed hostile carrier certificate supplied.

Theorem 17.2 (Gluon Color-Transport Carrier Closure). *If $\text{GaugeClaim}(x) \wedge \text{SameScope}_{D_{\text{gauge}}}(x)$ and x is a gluon-sector claim, then*

$$x \downarrow C_{\text{color-transport}} \vee \text{Rep}(x) \vee \text{Route}(x) \vee \text{Proj}(x) \\ \vee \text{Meas}(x) \vee \text{Update}(x) \vee \text{Burden}(x) \vee \text{FailPrim}(x).$$

Consequently,

$$\text{Survprim}_{D_{\text{gauge}}}(\text{Attr}_{\text{gluon}} \setminus C_{\text{color-transport}}) = 0$$

inside the scoped gauge comparison arena using the Yang–Mills endpoint support.

Status. Closed SM-2 color-transport carrier theorem using imported Yang–Mills endpoint support. It is not a reproof of the full Yang–Mills package inside SM-2 and not a numerical QCD phenomenology claim.

Gluon item	Status	Certificate note
Gluon field notation	representative	licensed by color-transport carrier.
Color index	representation/color-transport skin	not primitive color object by label alone.
Perturbative gluon line	route/calculation skin	valid perturbative display, not asymptotic primitive standing.
Jet	detector / projection route	evidence route, not isolated primitive standing.
Color-flow diagram	projection/route display	bookkeeping over color transport.
Wilson or loop display	obstruction or route display	licensed by curvature/holonomy and Yang–Mills endpoint support.
Hadronization route	route status	downstream route discipline.
Free isolated gluon claim	hostile falsifier branch	succeeds only with closed same-scope carrier certificate outside C_{coltrans} .

Criterion 17.3 (Free-gluon falsifier). A free-gluon falsifier must satisfy

$$\text{FreeGluon}(y) \wedge \text{SameScope}_{D_{\text{gauge}}}(y) \wedge \text{GCert}(y) \wedge y \not\downarrow C_{\text{color-transport}}.$$

Without those conditions, free-gluon language remains route, projection, burden, target update, or failed primitive.

18 Weak-Boson Standing: $W^\pm$ and Z

18.1 Submitted weak-boson claims

Weak-boson submissions include claims about $W^\pm$ and Z labels, weak field notation, weak mixing angle, pole masses, widths, decay channels, electroweak split, neutral/charged currents, longitudinal modes, Goldstone descriptions, Higgs vev explanations, and detector signatures.

18.2 Weak-boson carrier closure after equivalence completion

The weak-boson carrier is

$$C_{W/Z} = C_{\text{weak-chiral}} \wedge \text{GaugeVac}_D \wedge H_{\text{br/fix}}\text{-licensed mass-response standing.}$$

The L1 equivalence package closes the branch obligations charged-current collapse, EW/EM coupling bookkeeping, longitudinal/Goldstone equivalence, hypercharge/electric charge normalization, anomaly

freedom $\Leftrightarrow$ Yukawa realizability, absence of tree-level FCNC under admissible redescription, and equivalence-space closure. Together with A6, PSM6, SR6/SR7, the Higgs bridge/fixation arc, and mass-fixation support, these inputs license weak charged and neutral response standing after the electroweak split while keeping field notation, mixing coordinates, longitudinal/Goldstone descriptions, pole masses, widths, vev, and detector events derivative.

L1 / mass-fixation source	Imported payload used by SM-2	Source anchor
Charged-current collapse	Charged-current degree-of-freedom collapse used in the $W^\pm$ certificate.	Main equivalence-completion statement.
EW / EM coupling bookkeeping	Electroweak/electromagnetic bookkeeping bridge used in photon, W/Z, and weak-mixing status classification.	Bookkeeping-equivalence theorem.
Longitudinal and Goldstone equivalence	Longitudinal and Goldstone descriptions classified as equivalence or route statuses, not primitive W/Z grounds.	Equivalence theorem for longitudinal and Goldstone modes.
Hypercharge and electric-charge normalization	Charge-response normalization and electric-charge quantization support.	Charge-normalization theorem.
Anomaly / Yukawa realizability	Weak-sector compatibility between anomaly discipline and Yukawa/mass-response availability.	Anomaly/Yukawa equivalence theorem.
Tree-level FCNC absence	Neutral-current discipline supporting the Z-sector certificate.	FCNC redescription theorem.
Equivalence-space closure	No residual primitive equivalence-space escape.	Equivalence-space closure theorem.
Mass fixation	Weak-boson mass-response licensing through Higgs bridge/fixation support.	Mass-fixation theorem.

SM-2 uses the L1 package to close primitive-status questions about electroweak bookkeeping, longitudinal/Goldstone descriptions, charged/neutral current labels, and weak-boson mass-response licensing. It does not derive numerical electroweak parameters.

18.3 Weak-boson branch tree

For a same-scope weak-boson claim, the classifier branch is

- WeakBosonClaim(x) $\rightarrow$ Rep field notation or current label,
- $\rightarrow$ Meas pole mass, width, or detector event,
- $\rightarrow$ Trans/Mix weak mixing angle or basis convention,
- $\rightarrow$ Route decay channel or electroweak split description,
- $\rightarrow$ Carrier weak-chiral + gauge-vacuum + Higgs-bridge certificate,
- $\rightarrow$ Burden same-scope primitive asserted but certificate outstanding,
- $\rightarrow$ Update changed carrier inventory, boundary, or target,
- $\rightarrow$ FailPrim primitive asserted from skin alone.

Theorem 18.1 (Weak-Boson Weak-Chiral Carrier Closure). *If $\text{GaugeClaim}(x) \wedge \text{SameScope}_{D_{\text{gauge}}}(x)$ and x is a weak-boson claim, then*

$$x \downarrow (C_{\text{weak-chiral}} \wedge \text{GaugeVac}_D \wedge H_{\text{br/fix}}) \vee \text{Rep}(x) \vee \text{Meas}(x) \vee \text{Route}(x) \\ \vee \text{Proj}(x) \vee \text{Trans}(x) \vee \text{Update}(x) \vee \text{Burden}(x) \vee \text{FailPrim}(x).$$

Thus no residual same-scope primitive weak-boson standing survives outside the weak-chiral/gauge-vacuum/Higgs-bridge carrier package.

Status. *Closed SM-2 primitive-status theorem using A6/PSM6/SR6/SR7, L1 electroweak/equivalence support, Higgs bridge/fixation, and mass-fixation support. It is not a numerical derivation of M_W , M_Z , Γ_W , Γ_Z , θ_W , or the Higgs vev.*

18.4 W-boson certificate

Certificate field	$W^\pm$ certificate
Fixed target	D_{gauge} with weak-chiral response, charge ledger, anomaly ledger, gauge-vacuum compatibility, charged-current equivalence, longitudinal/Goldstone equivalence, and $H_{\text{br/fix}}$ fixed.
Carrier role	Charged weak-chiral response licensed by gauge-vacuum compatibility and Higgs bridge/fixation mass-response standing.
Source payload	A6/PSM6/SR6/SR7, charged-current collapse, EW/EM bookkeeping, longitudinal/Goldstone equivalence, anomaly/Yukawa compatibility, mass fixation, and Higgs bridge/fixation.
Licensed skins	$W^\pm$ field notation, charged-current route labels, longitudinal-mode descriptions, Goldstone-route language, pole mass, width, decay channels, detector events, and weak interaction event topologies.
Non-promotion rule	Charged-current label, longitudinal-mode description, Goldstone description, pole value, decay width, vev, or event signature does not select primitive W -identity.
Falsifier	Same-scope charged weak primitive outside weak-chiral/gauge-vacuum/Higgs-bridge carrier discipline with closed certificate.

18.5 Z-boson certificate

Certificate field	Z certificate
Fixed target	D_{gauge} with neutral weak response, charge ledger, anomaly ledger, gauge-vacuum compatibility, neutral-current discipline, EW/EM bookkeeping, and $H_{\text{br/fix}}$ fixed.
Carrier role	Neutral weak-chiral response after electroweak split, licensed by gauge-vacuum compatibility, Higgs bridge/fixation, and equivalence-space closure.
Source payload	A6/PSM6/SR6/SR7, EW/EM bookkeeping, longitudinal/Goldstone equivalence, absence of tree-level FCNC under admissible redescription, hypercharge/electric charge normalization, mass fixation, and Higgs bridge/fixation.
Licensed skins	Z field notation, neutral-current labels, weak mixing coordinate, longitudinal-mode descriptions, pole mass, width, decay route, and detector signature.
Non-promotion rule	Neutral-current label, mixing angle, pole value, width, longitudinal/Goldstone description, or detector route does not ground primitive Z -identity.
Falsifier	Same-scope neutral weak primitive outside weak-chiral/gauge-vacuum/Higgs-bridge carrier discipline with closed certificate.

18.6 Weak mixing angle, longitudinal/Goldstone status, pole mass, and width non-promotion

Weak-boson item	Status	Certificate note
W/Z field notation	representative	licensed by weak-chiral carrier.
Weak mixing angle	transport/mixing coordinate	not primitive transition identity.
Longitudinal mode	licensed route/equivalence status	not primitive identity by mode description alone.
Goldstone description	equivalence route	not primitive weak-boson content.
Pole mass	measurement witness / mass-response skin	witnesses mass-response standing.
Decay width	route/measurement witness	downstream route property.
Detector event	projection/measurement witness	valid event route.
Vev	Higgs-scale skin / downstream parameter	not primitive weak-boson ground.
Neutral/charged current label	route/representative	licensed by weak response.

Symmetry-breaking description	gauge-vacuum/Higgs bridge route	not gauge destruction or primitive selector.
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Criterion 18.2 (Weak-boson falsifier). A weak-boson falsifier would be a same-scope primitive weak-boson carrier outside weak-chiral/gauge-vacuum/Higgs-bridge discipline, with no target update, no derivative descent, and a closed carrier certificate.

19 Gauge Couplings and RG: Boundary Result for SM-4

19.1 Couplings are not gauge carriers

Gauge-coupling status is included here only as a boundary result. SM-2 proves the negative primitive-carrier result for coupling values; the positive transport, calibration, matching, scheme, and RG ledger theory belongs to SM-4.

Charge-response standing and coupling-strength calibration are distinct statuses. Photon standing uses charge-response standing; SM-2 does not derive or primitive-promote the numerical values e , g_i , α , or running $\alpha(\mu)$. The former is a carrier role; the latter is SM-4 transport/calibration material.

Proposition 19.1 (Gauge-Coupling Values Are Not Gauge Carriers).

$$\text{CouplingValue}(g_i) \wedge \neg \text{GCert}(g_i) \Rightarrow \neg \text{Primitive}_{\text{gauge}}(g_i).$$

Raw gauge-coupling values are transport/calibration/moduli coordinates unless additional invariant fixing, boundary selection, dynamical isolation, or empirical witness structure is supplied.

Coupling/RG object	SM-2 status	Handoff
g_1, g_2, g_3 values	transport/calibration ledger	SM-4 expands.
Electric charge value e	charge-response witness/calibration	SM-4 expands.
Running coupling	RG transport coordinate	SM-4 expands.
Unification plot	projection or transport display	BSM audit or target update if carrier inventory changes.
Fixed point claim	burden or target update unless isolation certificate supplied	SM-4/SM-6.

20 Gauge BSM and EFT Branch Certificates

20.1 Branch logic, not phenomenology catalogue

The following certificates are branch certificates, not single-model verdicts. Applying them to named models requires the model's declared field content, symmetry assumptions, coupling structure, empirical claim surface, and target declaration. SM-2 supplies the branch logic, not a catalogue of model by model phenomenological verdicts.

A gauge claim receives $\text{Ext}_{\text{gauge}}$ when it enriches or routes through the closed gauge carrier network without changing primitive carrier inventory, invariant bundle, source state, admissibility boundary, or comparison arena. It receives $\text{Update}_{\text{gauge}}$ when it changes any of those. Because C_{gauge} is closed in the scoped gauge arena, a same-scope primitive BSM survivor must supply an independent carrier certificate outside the closed gauge network.

Theorem 20.1 (Gauge BSM Branch-Certificate Theorem). *For a same-scope gauge-sector BSM claim b ,*

$$\rho_{\text{gauge}}(b) \in \{\text{Rep, Ext, Proj, Trans, Route, Update, Burden, FailPrim}\}$$

unless b supplies a same-scope closed carrier certificate outside the closed C_{gauge} network. In that case b is a genuine local gauge-sector falsifier.

20.2 Extra vector boson

Submitted claim: a new vector boson is a new primitive gauge carrier.

Certificate branch:

- If the claim is merely a field or particle label, it is **FailPrim** for primitive status.
- If it descends to a route, resonance, detector projection, or EFT display, it is a registered derivative status.
- If it preserves the gauge carrier package but adds a model surface, it is **Ext** or **Burden**.
- If it changes gauge-product inventory, representation ledger, charge lattice, anomaly ledger, or vacuum regime, it is **Update**.
- If it preserves same scope and supplies a closed carrier certificate outside C_{gauge} , it falsifies local gauge-sector closure.

20.3 Dark photon

Submitted claim: a dark photon is an additional primitive electromagnetic carrier.

Branch tests: charge-response preservation, kinetic-mixing status, projection/interface enrichment, hidden-sector carrier inventory, anomaly/vacuum compatibility, and target-arena change.

Expected branch:

$$\text{Proj} \vee \text{Ext} \vee \text{Burden} \vee \text{Update} \vee \text{FailPrim}.$$

Falsifier: independent same-scope electromagnetic primitive outside charge-response standing under fixed D_{gauge} .

20.4 Extra $U(1)$

Submitted claim: an extra $U(1)$ factor is a same-scope primitive gauge factor.

Tests: charge-lattice preservation, anomaly preservation, representation response, vacuum compatibility, coupling/normalization status, and hidden-load expansion. The likely branch is

$$\text{Update} \vee \text{Burden} \vee \text{FailPrim} \vee \text{Ext}.$$

A same-scope primitive extra $U(1)$ carrier is possible only if it preserves all existing ledgers while closing an independent carrier certificate.

20.5 Hidden non-Abelian gauge sector

Submitted claim: a hidden non-Abelian sector supplies an additional same-scope primitive gauge carrier.

Branch tests: whether the sector preserves the recovered D_{gauge} carrier inventory, whether its representation and charge/hidden-charge ledgers are part of the same comparison arena, whether anomaly and vacuum compatibility are preserved, whether confinement-like behavior is a route inside the fixed target or a new target regime, and whether the hidden sector introduces new source state or admissibility boundary.

Expected branch:

$$\text{Update} \vee \text{Burden} \vee \text{Ext} \vee \text{Proj} \vee \text{FailPrim}.$$

A hidden non-Abelian sector is a same-scope falsifier only if it preserves all gauge, representation, anomaly, vacuum, and boundary ledgers while closing an independent carrier certificate outside C_{gauge} .

20.6 Massive hidden vector / Stueckelberg-type vector

Submitted claim: a massive hidden vector is a primitive gauge-boson carrier inside the same SM gauge target.

Branch tests: whether the mass is a Higgs/Stueckelberg/model-surface route, whether the vector descends to projection or extension status, whether it changes carrier inventory, whether anomaly/vacuum compatibility is preserved, and whether a closed same-scope carrier certificate is supplied.

Expected branch:

$$\text{Ext} \vee \text{Proj} \vee \text{Trans} \vee \text{Update} \vee \text{Burden} \vee \text{FailPrim}.$$

A massive-vector field label or pole-mass value does not by itself supply primitive gauge standing.

20.7 Grand unification

Submitted claim: a unification group is the primitive gauge ground.

Tests: whether the proposal preserves D_{gauge} , whether the unification group is a transport/unification skin, whether the target/source/invariant bundle changes, and whether an independent carrier certificate is supplied.

Expected branch:

$$\text{Trans} \vee \text{Proj} \vee \text{Update} \vee \text{Burden}.$$

An RG crossing or unification plot does not generate primitive standing alone.

20.8 Gauge EFT anomaly

Submitted claim: a gauge-sector EFT coefficient or anomaly is primitive new gauge physics.

Expected branch:

$$\text{Proj} \vee \text{Trans} \vee \text{Route} \vee \text{Burden} \vee \text{Update}.$$

Falsifier: a same-scope EFT/gauge anomaly that cannot be represented as measurement, projection, route, or transport and supplies a closed primitive carrier certificate outside C_{gauge} .

21 Gauge Prediction Box

21.1 Target-preserving prediction and falsification

The prediction register in SM-2 has reality contact only because the target and comparison class are preserved. It is not an internal bookkeeping list. A predicted measurement, anomaly, route, or extension pressure is assessed as a prediction of the same gauge-sector claim only when the reference-bearing target in D_{gauge} remains fixed.

Proposition 21.1 (Reality-contact discipline for gauge predictions). *For every gauge prediction or falsification entry,*

$$\begin{aligned} & \text{Prediction}_{\text{gauge}}(x) \vee \text{Falsification}_{\text{gauge}}(x) \\ \Rightarrow & \text{KernelPreserved}_{\text{gauge}}(x) \wedge \text{TargetPreserved}(x) \wedge \text{ComparisonClassPreserved}(x). \end{aligned}$$

If the target changes, the result is a target update. If the target is preserved but the alleged primitive status does not close, the result is a registered status, burden, or failed primitive promotion rather than a same-scope primitive survivor.

21.2 Prediction-layer statement

SM-2's sector prediction is a primitive-status prediction:

$$\text{FutureG}(y) \Rightarrow (\exists \rho_{\text{gauge}}(y) \in \text{Status}_{\text{gauge}}) \vee \text{Failure}(G0).$$

Failure of the prediction is not hidden by ad hoc status expansion; it is recorded as pressure on the gauge classifier.

ID	Claim type	Expected disposition	Falsifier	Status
G0	new gauge-boson label	representative, extension, burden, target update, or failure	label alone supplies primitive standing	closed no-promotion
G1	photon deviation	measurement, route, projection, transport, burden, or update	same-scope photon primitive outside charge-response	closed SM-2 prediction
G2	free gluon	route/burden/cert.	primitive outside color transport	closed branch
G3	weak-boson deviation	measurement, route, projection, transport, burden, or update	weak primitive outside weak-chiral/Higgs-bridge carrier	closed SM-2 prediction

G4	new vector boson	representative, extension, projection, update, burden, or failure	field label alone supplies primitive standing	closed branch logic
G5	extra gauge factor	target update, burden, failed primitive, or licensed extension	same-scope primitive factor preserving all ledgers	closed branch logic
G6	coupling/RG crossing	transport, calibration, or projection	RG crossing creates primitive gauge standing	negative closed; positive theory to SM-4
G7	gauge anomaly	obstruction, burden, or update	anomaly supplies primitive standing without certificate	closed obstruction discipline
G8	gauge-vacuum deviation	route, burden, or update	vacuum coordinate primitive	closed route discipline
G9	hidden vector sector	target update, projection, extension, burden, or failure	hidden label same-scope primitive	closed branch logic
G10	GUT/unification claim	transport, projection, update, or burden	unification plot primitive	SM-4/SM-6 handoff

22 Final Scoped Gauge-Sector Primitive-Status Closure

Theorem 22.1 (Gauge-Sector Classification). *If $\text{GaugeClaim}(x) \wedge \text{SameScope}_{D_{\text{gauge}}}(x)$, then*

$$\rho_{\text{gauge}}(x) \in \text{Status}_{\text{gauge}} \setminus \{\text{Update}\}.$$

If x does not preserve same scope, it is classified under the target-update criterion.

Theorem 22.2 (No Primitive Gauge-Boson Skin Escape).

$$\text{GaugeSkin}(s) \wedge \neg \text{GCert}(s) \Rightarrow \neg \text{Primitive}_{\text{gauge}}(s).$$

In particular, field notation, gauge choice, polarization, propagator behavior, pole mass, width, detector event, jet signature, color-flow display, coupling value, RG coordinate, longitudinal/Goldstone description, and EFT coefficient do not self-promote into primitive gauge-boson standing.

Carrier certificate	Closure basis	Nonclaim boundary	Status
Gauge-order	R8 + A1	no ontic gauge-substance claim.	closed
Transport	A2	no numerical coupling derivation.	closed
Curv./hol.	A3	notation and Wilson displays remain skins.	closed

Charge-response / photon	A4 + PSM4 + SR4 + L1 charge and EW/EM support + Section 12	no derivation of e or α .	closed
Color-transport / gluon	A4 + PSM/SR + Yang–Mills endpoint package + Section 13	no QCD monograph reproof inside SM-2.	closed
Weak-chiral / W/Z	A6 + PSM6 + SR6/SR7 + L1 equivalence + Higgs/mass fixation + Section 14	no numerical EW parameter derivation.	closed
Anomaly	A5 + PSM5 + SR5 + L1 anomaly/Yukawa	no UV-completion ban.	closed
Gauge-vacuum	A6 + PSM6 + SR6 + Higgs/mass fixation	no primitive vev or vacuum coordinate.	closed

Lemma 22.3 (Certificate-to-Primitive-Status-Deletion Lemma).

$$\text{ClosedGCert}(C_{\text{gauge}}) \Rightarrow \text{Survprim}_{D_{\text{gauge}}}(\text{Attr}_G \setminus C_{\text{gauge}}) = 0.$$

If every primitive gauge role either descends to the listed carrier network or receives a registered derivative, burden, failure, or target-update status, no residual same-scope primitive gauge attribute remains outside C_{gauge} .

Theorem 22.4 (Gauge Carrier Certificate Closure).

$$\text{ClosedGCert}(C_{\text{gauge}}).$$

Proof spine. *The gauge-order, transport, curvature/holonomy, charge-response, color-transport, weak-chiral, anomaly, and gauge-vacuum certificates are closed by the dependency package listed in the ledger: SM-1 classifier discipline; R8; A1–A6; PSM1–PSM6; SR1–SR7; the Yang–Mills endpoint package; the L1 equivalence package; and Higgs/mass-fixation support. Each false primitive–field notation, gauge choice, polarization, propagator, pole mass, width, jet, color-flow display, coupling value, RG crossing, longitudinal/Goldstone description, or BSM label—is either licensed by the carrier network or assigned a registered nonprimitive status.*

Theorem 22.5 (Scoped Gauge-Sector Primitive-Status Deletion).

$$\text{Survprim}_{D_{\text{gauge}}}(\text{Attr}_G \setminus C_{\text{gauge}}) = 0.$$

This theorem deletes residual primitive-status survivors in the fixed same-scope gauge arena. It does not delete licensed gauge descriptions, empirical observables, numerical parameters, EFT displays, detector routes, or target-updating discoveries. It is not the global Standard Model endpoint, not numerical gauge-parameter closure, and not a catalogue-level phenomenological verdict for every named BSM model.

23 Minimal Hostile Gauge Reconstruction

23.1 Kernel-preserved hostile reconstruction

A hostile reconstruction has force only when it is a same-target reconstruction. It must preserve D_{gauge} , preserve the comparison class, preserve reference standing, and then supply an independent primitive role not already carried or licensed by the closed gauge package. Merely replacing the target, changing the admissibility rule, or appealing to a useful model surface is not a same-scope counterexample.

Criterion 23.1 (Faithful gauge counterexample). A faithful gauge counterexample must show all of the following: (i) kernel-preserved same-scope membership in D_{gauge} ; (ii) primitive role occupancy; (iii) non-derivativeness from the certified gauge carriers; (iv) nonmembership in the registered status atlas; and (v) target-preserving failure of the no-promotion theorem. If any item fails, the claim is a registered status, burden, failed primitive, or target update rather than a primitive survivor.

A genuine local counterexample must be stronger than a new label, anomaly, resonance, EFT coefficient, or route. Let y_{gauge}^* be a future gauge claim such that:

1. $\text{GaugeClaim}(y_{\text{gauge}}^*)$;
2. $\text{SameScope}_{D_{\text{gauge}}}(y_{\text{gauge}}^*)$;
3. $y_{\text{gauge}}^* \notin C_{\text{gauge}}$;
4. it is not a field label, gauge choice, polarization, propagator, coupling/RG coordinate, detector route, jet, color-flow display, EFT coordinate, or model name;
5. it is not a target update;
6. it defeats derivative descent;
7. it supplies a closed gauge carrier certificate.

Then

$$\text{Survprim}_{D_{\text{gauge}}}(y_{\text{gauge}}^* \setminus C_{\text{gauge}})$$

and SM-2's local closure fails. The classifier is not permitted to reclassify this object as target update unless the target-update criterion is satisfied, and it is not permitted to leave it as burden once the closed carrier certificate is supplied.

24 Referee Objection Lock

Objection	Response
Gauge bosons are already defined by fields.	Field representation is granted; primitive standing by field notation alone is denied.
Gauge symmetry is physical or ontic.	Gauge symmetry is standing-preserving redescription and transport organization, not primitive ontology.
This denies photons, gluons, W, and Z.	It preserves standard phenomenology as licensed representation, measurement, route, and projection.
The paper does not derive gauge couplings.	Correct. Coupling values are SM-4/parameter-sector targets, not SM-2 carrier standing.
Jets are evidence of gluons.	Jets are route or projection evidence for color-transport standing, not primitive isolated-gluon standing.
weak-boson pole masses define weak bosons.	Pole masses are measurement witnesses of weak-boson mass-response standing, not primitive identity grounds.
An extra gauge boson refutes this.	Only if it preserves same scope and supplies a carrier certificate outside C_{gauge} ; otherwise it is extension, burden, target update, or failure.

This is anti-BSM.

Gauge BSM claims are audit-admissible; they are not protected from primitive-status classification.

25 Lean 4 formalization target and public audit reference

The Lean 4 release is used here as a paper-shaped, proof-checked audit and certificate surface for the SM manuscript sequence. It records the endpoint carriers, declared hypotheses, theorem crosswalk inventory, and certificate interfaces in Lean; verifies that the stated closure theorems typecheck without `sorry`, `admit`, live local `axiom` declarations, or `unsafe` code in the audit surface; and ties the formal release to the DOI-bearing manuscript bundle. This is a substantive support layer: it gives a typed theorem registry, explicit hypothesis boundaries, machine-checked closure over declared certificate fields, an audit trail across SM-1–SM-6 and the endpoint paper, reproducibility checks, and DOI/release alignment. It should not be described as an independent reconstruction of the full Standard Model argument from foundational physics.

The generation-cardinality endpoint should be read at the governance-to-endpoint interface, not as a standalone phenomenological derivation inside the Standard Model sector. The mechanized surface represents the pre-physical admissibility constraints under which a non-degenerate physical presentation is admissible: open-arena discipline, no-selector and no-presentation-dependence constraints, role-occupancy closure, and boundary handoff. Within that framework, generation cardinality is a formal closure condition rather than a fit to particle content or an inference from arbitrary presentation-dependent structure. The Lean release audits that certificate surface and its dependencies; it does not reconstruct every analytic or phenomenological argument in the SM papers.

The public Lean-facing support package is the relevant SM ARC reference archive [46]: its README identifies the paper-shaped theorem surface, the locally received SM-1–SM-6 ARC layers, the generation-cardinality endpoint, the theorem-inventory layer, and the build/audit verifier. Its v0.1.0 release records the `GenerationCardinality` Lake target, carries SM-1 through SM-6 as local received ARC layers, preserves the generation-cardinality source artifacts, and reports build and focused axiom-audit checks over the endpoint anchors.

Lean audit item	Manuscript-facing role
Repository	<code>somamaley-ux/AASC-Standard-Model-Carrier-Exhaustion-Generation-Cardinality</code> , the public Lean 4 reference archive for the SM-1–SM-6 ARC carrier-exhaustion chain and the generation-cardinality endpoint.
Release DOI	v0.1.0 - Standard Model Carrier Exhaustion / Generation Cardinality. 10.5281/zenodo.21219417.
Primary theorem surface	<code>SM2.lean</code> and its imported <code>SM2/...</code> files, read with <code>SM1.lean</code> , <code>SM3.lean</code> – <code>SM6.lean</code> , and <code>GenerationCardinality.lean</code> as received support surfaces.
Axiom audit command	Repository verifier <code>scripts/check-generation-cardinality-audit.ps1</code> with the <code>-SkipLakeUpdate</code> option; the verifier builds the endpoint target, scans for live <code>axiom</code> , <code>sorry</code> , <code>admit</code> , and <code>unsafe</code> declarations, and runs focused axiom checks over endpoint anchors.
Build target	<code>GenerationCardinality</code> , with aggregate imports through <code>SM1.lean</code> – <code>SM6.lean</code> and <code>GenerationCardinality.lean</code> .

Audit meaning	The public release machine-checks the governance-to-endpoint certificate surface over declared carriers, received ARC layers, and explicit manuscript hypotheses. Once the pre-physical admissibility layer is accepted, the endpoint predicate is a formal closure condition, not a phenomenological fit or downstream presentation selector.
Certificate/interface value	The layer is not merely bookkeeping: it supplies a typed theorem registry, explicit hypothesis boundaries, machine-checked closure over declared certificate fields, an SM-1–SM-6 audit crosswalk, reproducibility checks, and DOI/release alignment.
Boundary	The cardinality endpoint is not a standalone SM-sector derivation from particle dynamics; it is a governance-level endpoint that makes explicit the admissibility, no-selector, open-arena, role-occupancy, and presentation-independence constraints under which later SM interpretations are meaningful. The Lean release audits this certificate/interface boundary and its dependencies; it does not reconstruct every analytic or phenomenological argument of SM-1–SM-6.

A manuscript-faithful Lean route for SM-2 represents:

1. fixed scopes and submitted-claim or candidate-record sets as explicit carriers;
2. carrier packages, registered-status packages, and received sector packages as typed structures rather than implicit background metaphysics;
3. same-scope and target-update predicates as explicit boundary conditions;
4. derivative-skin, representative, route, projection, transport, burden, failure, and extension statuses as typed predicates or classifiers;
5. certificate closure and no-promotion principles as theorems over declared hypotheses, not as global axioms;
6. finite branch exhaustions for the relevant same-scope arena;
7. the handoff to SM-6, or from SM-6 to the generation-cardinality endpoint, as an explicit typed support relation.

For SM-2, the paper-specific audit surface is the gauge-sector status package: gauge-order carrier, connection/transport and curvature/holonomy status, charge-response carrier, color-transport carrier, weak-chiral carrier, anomaly and vacuum-compatibility branches, photon/gluon/weak-boson certificates, gauge-coupling non-carrier status, BSM gauge branches, and scoped gauge-sector primitive deletion.

A manuscript-faithful formalization over the declared carriers and hypotheses requires no custom axiom at a load-bearing node. If a proposed audit file relies on `axiom`, `sorry`, `admit`, `unsafe`, or an opaque proof-carrying placeholder at such a node, it is not counted as formal support for this manuscript. Conversely, a clean axiom audit is not a stand-alone validation of the physical hypotheses encoded as record fields; it is a proof-checked governance-to-endpoint certificate boundary whose value is typed traceability, explicit assumptions, reproducible build discipline, and DOI-aligned auditability.

26 Handoff to Later Papers

26.1 Handoff to SM-3

SM-2 supplies color-transport discipline, no primitive isolated-gluon skin, and the gauge side of quark/gluon interaction standing. SM-3 handles quark matter standing and confinement-compatible quark carrier closure.

26.2 Handoff to SM-4

SM-2 supplies the gauge-coupling non-carrier proposition and treats coupling/RG displays as transport/calibration ledgers. SM-4 handles charge, coupling, matching, scheme, and RG transport in full.

26.3 Handoff to SM-5

SM-2 supplies weak-chiral gauge response and the gauge side of mixing/transition standing. SM-5 handles CKM/PMNS, phase, basis, and generation-cardinality issues.

26.4 Handoff to SM-6

SM-2 hands forward the closed C_{gauge} package, the gauge skin atlas, the gauge BSM branch certificates, the scoped gauge-sector primitive-status deletion theorem, and the gauge prediction box. SM-6 uses them in the final assembly

$$C_{\text{SM}} = C_{\text{gauge}} \cup C_{\text{matter}} \cup C_{\text{transport}} \cup C_{\text{flavor}} \cup C_H \cup C_{\text{projection}}.$$

27 Conclusion

The kernel-status strengthening is part of the theorem, not a commentary layer. The closure claimed here is primitive-status closure under the minimal non-degeneracy conditions required for gauge claims to remain the same standing-bearing physical claims across measurement, prediction, comparison, and attempted falsification.

SM-2 closes the local gauge-sector primitive-status result under the declared same-scope gauge arena. The closed carrier package is

$$C_{\text{gauge}} = \{C_{\text{gauge-order}}, C_{\text{transport}}, C_{\text{curv/hol}}, C_{\text{charge-response}}, \\ C_{\text{color-transport}}, C_{\text{weak-chiral}}, C_{\text{anomaly}}, C_{\text{gauge-vacuum}}\}.$$

Photon standing descends to charge-response and unbroken electromagnetic transport. Gluon standing descends to non-Abelian color transport supported by the Yang–Mills endpoint package. $W^\pm$ and Z standing descends to weak-chiral/gauge-vacuum/Higgs-bridge licensed response, with L1 equivalence and mass-fixation support blocking primitive promotion of mixing, longitudinal/Goldstone, vev, pole-mass, and width descriptions.

SM-2 therefore proves

$$\text{Survprim}_{D_{\text{gauge}}}(\text{Attr}_G \setminus C_{\text{gauge}}) = 0.$$

This is not empirical no-new-physics, numerical gauge-parameter closure, or global Standard Model primitive exhaustion. Numerical gauge couplings, running values, weak mixing, W/Z pole masses, widths, and weak-scale values remain downstream parameter, witness, or transport targets. Gauge-sector BSM claims remain audit-admissible: they become representatives, measurements, routes, projections, transport ledgers, extensions, target updates, burdens, failures, or genuine same-scope falsifiers according to the stated certificate conditions.

The output handed to SM-6 is the closed gauge carrier network, the gauge skin atlas, the gauge prediction register, and the scoped local primitive-status deletion theorem. SM-6 may use this package as the gauge-sector component of the final Standard Model carrier-exhaustion synthesis.

A1 Gauge Dependency Ledger

This appendix records the exact dependency posture for SM-2. The table is not a bibliography substitute. It is an audit map: each source is used for a typed payload, and no source is asked to prove more than its declared role.

Dependency	Imported payload used by SM-2	Nonclaim preserved in SM-2
SM-1 classifier	carrier/skin grammar; target-update criterion; burden state; no-promotion theorem; prediction-register form	does not prove gauge carriers by itself.
Standalone SM spine	recovered Σ_{SM} , including G_{SM} , representation-response skeleton, charge lattice, anomaly ledger, vacuum-compatible residual form, and $H_{\text{br/fix}}$	does not close numerical couplings or all parameter values.
Higgs package	bridge/fixation carrier and licensed Higgs-sector skins	does not make vev or scalar potential primitive.
R8	boundary-trace gauge role	does not supply a transition groupoid or Standard Model gauge product at this layer.
A1	boundary-fixed transition groupoid	does not supply connection, curvature, representation, anomaly, or vacuum compatibility at this layer.
A2	standing-preserving transport comparison	does not make potentials primitive.
A3	curvature/holonomy obstruction carrier	does not make field-strength symbols primitive.
A4	representation and charge-witness rigidity	does not make representation labels or charge numbers primitive.
A5	anomaly as standing-failure obstruction	does not treat anomaly equations as primitive objects.
A6	gauge-vacuum compatibility	does not derive Higgs numerical values or electroweak parameter table.

SR1	simultaneous constraint collapse	does not select by empirical convenience.
SR2	product-sector rigidity	does not make group labels primitive.
SR3	representation-response rigidity	does not make representation notation primitive.
SR4	Abelian charge-ratio rigidity	does not derive every charge formula by itself.
SR5	anomaly-ledger rigidity	does not ban UV completion or target updates.
SR6	residual/vacuum-sector rigidity	does not solve cosmological constant or numerical vacuum matching.
SR7	minimal SM-like survivor profile in the fixed comparison regime	does not derive Higgs dynamics, Yukawa matrices, gauge coupling values, or mass generation mechanics.
Gauge non-onticity paper	gauge symmetry as representational bookkeeping rather than primitive ontology	does not deny gauge theory's utility.
Gauge coupling/moduli papers	coupling values as moduli/transport/calibration statuses	does not prove numerical gauge-coupling closure.

A1.1 Core gauge-realization chain and full SM-2 closure stack

The core gauge-realization chain used by SM-2 is

$$R8 \Rightarrow A1 \Rightarrow A2 \Rightarrow A3 \Rightarrow A4 \Rightarrow A5 \Rightarrow A6 \Rightarrow SR1-SR7.$$

At each arrow a new carrier or obstruction ledger is added. This core route is not a list of independent assumptions: boundary trace fixes gauge role; transition organization makes local presentation comparison compositional; connection-like transport makes comparison stable; curvature/holonomy records loop residue; representation/charge attaches gauge structure to response; anomaly and vacuum compatibility fail-close non-surviving candidates; SR1–SR7 compress the survivor ledgers into the recovered Standard Model gauge skeleton.

The full SM-2 closure stack includes the bridge, color, electroweak, and Higgs/mass-fixation sources required to close the gauge-boson carrier certificates:

$$R8 \Rightarrow A_{1-6} \Rightarrow PSM_{1-6} \Rightarrow SR_{1-7} \Rightarrow (YM_{\text{endpoint}}, L1_{\text{EW}}, H_{\text{br/fix}}) \Rightarrow \text{ClosedGCert}(C_{\text{gauge}}).$$

Thus the R8–A1–A6–SR1–SR7 display is the core gauge-realization route; the full primitive-status closure theorem also uses PSM1–PSM6, the Yang–Mills endpoint package, the L1 equivalence package, and Higgs/mass-fixation support.

A1.2 Dependency availability and proof-load status

Dependency family	Availability status	SM-2 dependency status	Trust/audit note
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SM-1 classifier paper	project archive PDF/source	load-bearing	classifier source for carrier/skin, target-update, burden, prediction, and hostile-falsifier discipline.
Standalone SM spine	project archive PDF/source	load-bearing structural payload	supplies recovered Σ_{SM} and numerical nonclaims.
Higgs package	project archive PDF/source	load-bearing for W/Z	bridge/fixation carrier support.
R8 and SR1–SR7	project archive PDF/source	load-bearing	boundary-trace and realization-rigidity support.
A1–A6	project archive PDF/source	load-bearing	gauge-realization ladder support.
PSM1–PSM6	project archive PDF/source	load-bearing bridge	Lie, product, representation, charge, anomaly, and residual/vacuum bridge support.
Yang–Mills endpoint package	project archive PDF/source	load-bearing for color	color-transport/gluon carrier support; project archive records available PDF/source status.
L1 equivalence package	project archive PDF/source	load-bearing for electroweak and charge	EW/EM, charged-current, longitudinal/Goldstone, hypercharge/charge, anomaly/Yukawa, FCNC, and equivalence-space support.
Gauge coupling and gauge ontology papers	project archive PDF/source	load-bearing boundary/orienta- tion	supports non-ontic gauge posture and coupling-moduli non-promotion.
External QFT/EFT literature	conventional orientation only	orientation	fixes standard vocabulary; does not supply AASC proof authority.

A2 Detailed Source Payload Ledger

The source anchors in this appendix are intentionally coarse where support papers use named endpoint certificates rather than uniform theorem numbering. For external audit, the indicated theorem, certificate, or closure section is the first location to inspect in the corresponding support paper.

A2.1 R8 payload

R8 supplies the first gauge-facing role. Its output is not a group and not a field; it is boundary-trace-fixed presentation equivalence. The role is necessary because local physical presentation must preserve boundary-detectable standing under admissible redescription. SM-2 uses R8 to justify the first carrier C_{gord} , not to derive G_{SM} directly.

A2.2 A1 payload

A1 turns the R8 gauge role into $\text{Gpd}_\partial(D)$. The transition groupoid records local-domain objects, restriction compatibility, identities, inverses where admissible, and composition of standing-preserving transition maps. The false primitive deleted by A1 is the local gauge chart as primitive object.

A2.3 A2 payload

A2 adds the connection-like transport comparison $\nabla_{\partial D}$. It is a carrier of standing-preserving comparison, not primitive potential ontology. SM-2 uses it to classify potentials, covariant-derivative notation, and connection coefficients as licensed transport representatives.

A2.4 A3 payload

A3 adds $\Omega_{\partial D}$, the curvature/holonomy obstruction role. A loop residue carries standing only when it is boundary-detectable, quotient-stable, same-domain, and transport-witnessed. SM-2 uses this to classify field-strength notation and loop displays as representatives of obstruction standing.

A2.5 A4 payload

A4 adds representation-response and charge-witness rigidity. Gauge structure becomes physically loaded only through admissible response roles. SM-2 uses this to prepare the photon theorem and to block primitive promotion of representation labels, charge numbers, currents, and detector charge signatures.

A2.6 A5 payload

A5 makes anomaly obstruction a standing-failure ledger. The anomaly test is not optional algebraic tidiness. A same-scope gauge realization that fails the anomaly ledger cannot be a survivor under fixed gauge standing. SM-2 uses this to discipline extra gauge factors and anomalous BSM branches.

A2.7 A6 payload

A6 adds gauge-vacuum compatibility. Gauge standing must preserve vacuum support and residual-sector conditions. SM-2 uses this to prepare weak-boson standing and to block primitive promotion of vacuum coordinates, symmetry-breaking slogans, or vev displays.

A2.8 SR1-SR7 payload

SR1-SR7 compress the gauge arena after the A-ladder supplies the admissible components. SR1 prevents free factorization of product, representation, charge, anomaly, vacuum, residual, transport, and deformation ledgers. SR2-SR6 close product, representation, charge, anomaly, and residual/vacuum projections. SR7 assembles the joint survivor into a minimal Standard-Model-like role profile in its comparison regime. SM-2 uses this as gauge-product survivor support, not as numerical parameter-table closure.

A3 Full Gauge Carrier Register

Carrier	Primitive role	Certificate obligations	Licensed skins	Falsifier
C_{gord}	gauge-order standing	boundary trace, transition closure, redescription invariance	gauge charts, local presentations, transition notation	same-scope gauge order outside boundary-fixed transition organization.
$\text{Gpd}_{\partial}(D)$	transition composition	identities, local inverses, restriction compatibility, composition	transition functions, patches, local groupoid notation	standing-preserving transitions that cannot be groupoid-organized.
$\nabla_{\partial D}$	transport comparison	path comparison, quotient stability, boundary compatibility	potentials, connection coefficients, covariant-derivative notation	primitive potential standing independent of transport comparison.
$\Omega_{\partial D}$	loop obstruction	boundary detectability, loop residue, same-domain transport witness	field strength, holonomy display, Wilson-loop-like display	obstruction standing outside transport residue.
$\text{Rep}_{\partial D}$	representation response	response role, quotient-stable carrier action, boundary witness	representation labels, generator charts, multiplet notation	representation standing by label alone.
$\text{ChWit}_{\partial D}$	charge witness	representation attachment, charge-response witness, normalization discipline	charge numbers, currents, detector charge readouts	charge primitive outside response witness.
$\text{Anom}_{\partial D}$	obstruction ledger	anomaly neutrality, fail-closed standing, repair exclusion	trace sums, algebraic equations, regularization displays	anomalous same-scope standing preserving all ledgers.
GaugeVac_D	vacuum compatibility	residual support, broken/unbroken route discipline, Higgs bridge boundary	phase descriptions, vacuum branch coordinates, vev display	weak gauge standing outside vacuum compatibility.
C_{γ}	unbroken electromagnetic charge-response transport	charge response, unbroken route, representation/witness preservation	photon field, polarization, Coulomb display, detector events	same-scope photon primitive outside charge-response.

C_g	color transport	non-Abelian product role, color response, confinement-compatible route discipline	gluon field, color labels, perturbative lines, jets	free same-scope primitive gluon certificate.
$C_{W/Z}$	weak-chiral response	weak transition, gauge-vacuum compatibility, Higgs bridge/fixation mass response	W/Z fields, pole masses, widths, weak currents	weak primitive outside weak-chiral/gauge-vacuum/Higgs carrier.

A4 Full Gauge Skin Atlas

The following atlas is used both for local proofs and for future gauge-claim audit.

Skin	Dependence test	Licensed status	Non-promotion rule
Gauge potential	gauge and chart dependence	transport representative	does not self-ground gauge standing.
Connection coefficients	chart/coordinate dependence	local transport skin	descend to $\nabla_{\partial D}$.
Gauge fixing	gauge-choice dependence	calculation route	no primitive standing by gauge choice.
Ghost fields	gauge-fixing formalism	calculational skin	no primitive physical gauge carrier.
Propagator	gauge, scheme, and perturbative route	calculation/route skin	no primitive identity by propagator form.
Polarization vector	basis and measurement-route dependence	representative or measurement skin	no primitive standing by polarization.
Field strength	representative of obstruction	curvature/holonomy display	descends to $\Omega_{\partial D}$.
Generator chart	chart/basis dependence	algebraic notation	no primitive standing by generator label.
Group label	product-sector presentation	product representative	needs product carrier certificate.
Charge value	normalization and representation attachment	charge-witness display	no primitive charge object by number.
Current	route/measurement dependence	charge-response witness	witnesses standing; does not create it.
Coupling value	calibration, scale, scheme	transport ledger	no primitive gauge carrier by value.
RG flow	scheme and scale route	transport ledger	no primitive standing by running.

Unification plot	projection and scale route	transport or projection display	crossing is not primitive carrier.
Pole mass	measurement and mass-response route	measurement witness	not primitive boson identity.
Width	decay-route dependence	route witness	not primitive identity.
Jet	detector/projection route	color-transport evidence route	not isolated primitive gluon.
Color-flow diagram	bookkeeping or projection dependence	route display	no primitive color object by diagram.
Detector event	apparatus or projection dependence	measurement witness	event does not generate carrier.
EFT coefficient	basis, matching, or scheme dependence	projection or transport status	no primitive standing without carrier pressure.
BSM vector label	model-name dependence	branch-certificate input	label alone fails primitive promotion.

A5 Complete Gauge Prediction Register

ID	Claim type	Expected classification	Excluded classification / falsifier	Status
G0	new gauge-boson label	representative, extension, burden, update, or failure	label alone primitive	closed no-promotion.
G1	photon-sector deviation	measurement, route, projection, transport, burden, or update	photon primitive outside charge-response	closed SM-2 theorem.
G2	free gluon	burden unless certificate closes	free same-scope primitive gluon	closed falsifier branch.
G3	weak-boson deviation	measurement, route, projection, transport, burden, or update	weak primitive outside weak carrier package	closed SM-2 theorem.
G4	new vector boson	representative, extension, projection, update, burden, or failure	field label primitive	branch certificate.
G5	extra gauge factor	target update, burden, failed primitive, or licensed extension	same-scope primitive gauge factor preserving ledgers	branch certificate.
G6	coupling/RG crossing	transport, calibration, or projection	crossing creates primitive gauge standing	SM-4 handoff.

G7	gauge anomaly	obstruction, burden, or update	anomaly primitive without carrier certificate	closed obstruction discipline.
G8	gauge-vacuum deviation	route, burden, or update	vacuum coordinate primitive	Higgs/SM-6 handoff.
G9	hidden vector sector	target update, projection, extension, burden, or failure	hidden label same-scope primitive	BSM branch.
G10	GUT/unification claim	transport, projection, update, or burden	unification plot primitive	SM-4 and SM-6 handoff.

A6 Expanded Worked Classification Certificates

A6.1 Photon certificate

Audit field	Certificate content
Submitted claim	The photon is primitive because it is the electromagnetic field quantum.
Fixed target	D_{gauge} with recovered gauge product, charge lattice, anomaly ledger, and unbroken electromagnetic charge-response route.
Carrier candidate	$C_\gamma = C_{\text{chresp}} \wedge C_{\text{unbroken-EM transport}}$.
Skin tests	A_μ , $F_{\mu\nu}$, polarization, detector events, and Coulomb displays are gauge/route or projection dependent.
Descent	Field and measurement language descends to charge-response transport and curvature/transport carriers.
Burden/falsifier	Same-scope primitive photon carrier outside charge-response with closed certificate.
Verdict	Carrier standing is charge-response transport; familiar photon language is licensed representative/witness/route.

A6.2 Gluon certificate

Audit field	Certificate content
Submitted claim	A gluon is primitive as a free particle-like gauge boson.
Fixed target	D_{gauge} with non-Abelian product role, representation response, anomaly ledger, and vacuum compatibility.
Carrier candidate	$C_g = C_{\text{coltrans}}$.
Skin tests	Perturbative lines, color indices, jets, and color-flow diagrams are route or projection/calculation dependent.
Descent	Gluon language descends to non-asymptotic color-transport standing.

Burden/falsifier	Free isolated gluon with same-scope primitive standing outside C_{coltrans} .
Verdict	Carrier plus derivative route or projection skins; free-gluon assertion remains burden unless certificate closes.

A6.3 W boson certificate

Audit field	Certificate content
Submitted claim	The W boson is primitive because its field, pole mass, and charged-current decays are observed.
Fixed target	D_{gauge} with weak-chiral role, gauge-vacuum compatibility, and Higgs bridge/fixation boundary.
Carrier candidate	$C_{\text{weakch}} \wedge \mathbf{GaugeVac}_D \wedge H_{\text{br/fix}}$ -licensed mass response.
Skin tests	W-field notation, pole mass, width, charged-current label, and detector signature are representatives or witnesses.
Descent	All lawful W language descends to weak-chiral/gauge-vacuum/Higgs bridge discipline.
Burden/falsifier	Same-scope primitive weak carrier outside this package.
Verdict	Weak-boson carrier standing plus measurement/route skins.

A6.4 Z boson certificate

Audit field	Certificate content
Submitted claim	The Z boson is primitive because neutral-current signatures and pole mass identify it.
Fixed target	D_{gauge} with weak-chiral role and electroweak split licensed by gauge-vacuum/Higgs bridge discipline.
Carrier candidate	$C_{\text{weakch}} \wedge \mathbf{GaugeVac}_D \wedge H_{\text{br/fix}}$ -licensed neutral weak response.
Skin tests	Z notation, pole mass, width, weak mixing angle, and neutral-current route are derivative statuses.
Descent	Neutral weak standing descends to weak-chiral carrier discipline.
Burden/falsifier	Primitive neutral weak carrier outside weak-chiral package under same scope.
Verdict	Carrier standing plus licensed weak-sector skins.

A6.5 Gauge potential certificate

A gauge potential is not rejected. It is one of the most useful local representatives. Its status is nevertheless derivative. The potential varies under admissible gauge redescription and its standing is inherited from $\nabla_{\partial D}$. A primitive-potential claim must supply redescription-invariant role closure independent of the transport carrier; absent that, the potential remains a licensed transport representative.

A6.6 Jet certificate

A jet is an evidence route for color-transport standing. It is detector-facing, projection-dependent, and hadronization-route dependent. Its success as evidence for color transport is not a primitive-isolated-gluon result. A jet falsifies SM-2 only if the associated claim supplies same-scope primitive gluon standing outside C_{coltrans} .

A6.7 Dark photon certificate

A dark photon claim is branch-tested. Kinetic-mixing language is projection or transport status. A hidden sector is target update if it changes carrier inventory. A detector excess is measurement/route evidence. A same-scope primitive electromagnetic carrier outside charge-response is a burden until a certificate closes and a falsifier if it closes.

A6.8 Gauge EFT anomaly certificate

An EFT coefficient is a low-energy projection and matching coordinate unless it supplies carrier pressure. The audit tests operator-basis dependence, matching route, scheme transport, measurement surface, and target update. It becomes a serious burden only if projection, transport, route, and measurement statuses are exhausted under fixed D_{gauge} .

A7 Gauge BSM Branch Matrix

Claim type	Same-scope branch	Target-update branch	Primitive falsifier branch
Extra vector boson	extension, projection, or burden if ledgers preserved	hidden inventory or new invariant bundle	carrier outside C_{gauge} preserving all ledgers.
Dark photon	kinetic mixing or projection-interface enrichment	hidden-sector carrier inventory	independent EM primitive outside C_{chresp} .
Extra $U(1)$	burden if charge/anomaly/vacuum ledgers preserved but certificate open	changed charge lattice or anomaly ledger	same-scope Abelian carrier outside recovered product.
GUT group	transport or unification projection	larger-domain unification target	primitive group carrier preserving D_{gauge} without target change.
Gauge EFT anomaly	projection, transport, or route burden	changed effective regime or invariant bundle	carrier-pressure outside C_{gauge} with certificate.
Composite vector	route/model surface burden	changed substructure target	primitive same-scope composite gauge carrier.
Massive hidden vector	detector projection or extension	hidden-sector target	same-scope primitive vector carrier.

A8 Proof Obligation Ledger

Theorem	Obligations discharged	Remaining boundary
Boundary-trace gauge role	boundary trace, presentation equivalence, realized physical interior	no group/product/coupling derivation.
Transition carrier	identities, composition, restriction compatibility	no connection/curvature yet.
Connection carrier	standing-preserving transport comparison	no primitive potential.
Curvature carrier	loop residue, boundary detectability, quotient stability	no primitive field-strength symbol.
Representation/charge carrier	response role, charge witness, normalization discipline	no primitive charge number.
Anomaly obstruction	standing-failure ledger, anomaly-neutral survivor test	no UV-completion ban.
Gauge-vacuum compatibility	residual/vacuum support, bridge boundary	no Higgs numerical closure.
Photon carrier	unbroken charge response and EM transport	no primitive event/polarization.
Gluon carrier	non-Abelian color transport and confinement-compatible route discipline	free-gluon burden remains falsifiable.
Weak carrier	weak-chiral response plus Higgs bridge/fixation	no primitive pole mass or vev.
Gauge coupling non-carrier	moduli/transport status	SM-4 handles full coupling/RG closure.
Gauge primitive-status deletion	all above local certificates	global SM endpoint remains SM-6.

A9 Lean 4 audit surface ledger

The gauge-sector Lean-facing surface is the public audit reference described in Section 25. The local ledger below records manuscript anchors and their Lean-facing roles without introducing an axiom-bearing declaration block.

Gauge audit anchor	Lean-facing role
Gauge order	carrier status for gauge-order standing and lawful redescription
Photon standing	charge-response and unbroken electromagnetic transport certificate
Gluon standing	non-Abelian color-transport certificate with confinement-compatible route discipline
Weak-boson standing	weak-chiral response and Higgs-bridge/fixation compatibility

Derivative gauge skins	field coordinate, gauge fixing, ghost, polarization, propagator, pole-mass, and coupling displays as non-promoting statuses
BSM gauge branches	extension, burden, target update, failure, or same-scope hostile certificate branch
Scoped deletion	theorem-level closure of $\text{Survprim}_{D_{\text{gauge}}}(\text{Attr}_G \setminus C_{\text{gauge}}) = 0$ under the declared gauge certificates

A10 Handoff Certificate to SM-6

SM-6 receives from SM-2 the following listed local package under the declared gauge certificates:

1. C_{gauge} , the gauge carrier network.
2. The gauge skin atlas and no-promotion theorem.
3. Photon, gluon, and weak-boson standing certificates.
4. Gauge BSM branch certificates.
5. Gauge prediction register G0-G10.
6. Local primitive-status deletion $\text{Survprim}_{D_{\text{gauge}}}(\text{Attr}_G \setminus C_{\text{gauge}}) = 0$ under the gauge-sector certificates.

SM-6 must not reprove all SM-2 details. It should cite the local closure and use it as the gauge-sector component of C_{SM} . If SM-3, SM-4, SM-5, and the Higgs/projection-interface closures also supply their local deletion results, SM-6 may then assemble the global endpoint.

A11 Submission-Facing Reading Guide

A reader can audit SM-2 in four passes. First, verify that SM-2 does not make gauge symmetry ontic and does not demote standard gauge-theory language. Second, verify the R8/A1-A6 ladder and the SR1-SR7 survivor payload. Third, inspect the photon, gluon, and weak-boson certificates. Fourth, inspect the hostile reconstruction and BSM branch certificates. If a proposed objection does not preserve D_{gauge} , defeat derivative descent, and close a carrier certificate outside C_{gauge} , it does not defeat SM-2.

A12 Detailed Source Payload Compression

This appendix records the source payload used by SM-2 at a level sufficient for manuscript review. The compression rule is strict: a source is compressed only to an output that is explicitly used by a downstream gauge-sector classification. The purpose is not to reprint the source papers. The purpose is to identify the exact theorem payload used, the false primitive it removes, and the local carrier role it supports.

A12.1 R8: boundary-trace gauge role

R8 supplies the point at which gauge structure first becomes a physical role. The payload is that a realized physical interior with boundary traces cannot treat local presentations as independent primitive descriptions if those presentations must preserve field-carrier mediation, particle-persistence identity,

quotient comparison, bounded transport, and boundary witnesses. Gauge structure is therefore first admitted as boundary-trace-fixed presentation equivalence. SM-2 uses this to block two mistakes: treating gauge symmetry as ontic and treating one gauge representative as the primitive occupant.

The false primitive deleted by R8 is the idea that gauge structure is a free-standing physical substance prior to admissible redescription. The carrier it supports is $C_{\text{gauge-order}}$. The falsifier would be a same-scope gauge role preserving the physical interior and boundary witnesses while avoiding boundary-trace-fixed presentation equivalence. No such object is supplied in the declared dependency package.

A12.2 A1: transition groupoid payload

A1 sharpens the minimal gauge role into local/restriction transition organization. Once admitted local gauge presentations exist, the transitions between them cannot be arbitrary pairwise identifications. Identity transitions are forced by same-presentation standing preservation; inverse transitions are forced by reversible role equivalence; partial composition is forced by composition-admissible interfaces; associativity is forced because bracketing-dependent role output would either be quotient-collapsed skin or standing failure; triple-interface coherence is required because path-dependent quotient realization destroys admissible fixation.

SM-2 uses A1 to license local gauges, transition functions, patching data, and generator charts as valid skins of transition organization. The false primitive deleted is local gauge presentation as primitive physical occupant. The carrier is $\text{Gpd}_{\partial}(D)$. A future gauge-sector claim defeats this step only if it supplies standing-preserving local presentation freedom that is not organized by the boundary-fixed groupoid role while still preserving same scope.

A12.3 A2: connection transport payload

A2 adds transport comparison. The transition groupoid says how presentations overlap; it does not by itself compare carrier content along typed continuation paths. A2 proves that stable physical gauge realization requires a standing-preserving transport-comparison structure. Raw representative comparison, global gauge fixing, arbitrary path assignment, or isolated overlap transitions fail because they import hidden selectors, destroy quotient stability, fail bounded-transport compatibility, or make transported standing path-dependent without admissible witness.

SM-2 uses A2 to classify gauge potentials and connection coefficients. A potential is lawful as a transport representative; it is not primitive gauge ontology. The false primitive deleted is A_{μ} as primitive physical object. The carrier is $\nabla_{\partial D}$, the boundary-fixed standing-preserving comparison assignment.

A12.4 A3: curvature/holonomy payload

A3 asks what counts as admissible nontrivial residue around local continuation loops once transport comparison exists. It proves that loop residue has standing only when it is boundary-detectable, quotient-stable, same-domain, transport-witnessed, gauge-compatible, and standing-bearing. Raw path dependence, arbitrary loop labels, hidden endpoint closure, global representative selection, boundary-invisible residue, non-composable transport, role-changing loop output, causal-reversal import, and cross-domain drift fail as curvature-role residue.

SM-2 uses A3 to license field-strength notation and holonomy displays as representatives of obstruction standing. The false primitive deleted is a field-strength formula as primitive object. The carrier is $\Omega_{\partial D}$, the transport-holonomy obstruction role.

A12.5 A4: representation and charge-witness payload

A4 supplies the response layer. A candidate representation response is admissible only if its response to gauge transitions, transport comparisons, and loop residues preserves field-carrier mediation, particle-persistence identity, quotient comparison, bounded transport witnesses, boundary/classification witnesses, and same-domain standing. Representation labels that fail transport compatibility, depend on presentation choice, alter particle persistence, erase loop-residue standing, require hidden endpoint selection, or introduce unwitnessed internal multiplicity are inadmissible as physical representation-response structure.

The charge result is equally important: a charge is a boundary-detectable quotient-stable witness of representation response. Raw charge labels, normalizations, and weights collapse unless the admitted witness ledger distinguishes them. SM-2 uses this to ground photon standing and charge-response standing while blocking charge-number primitive promotion.

A12.6 A5: anomaly-obstruction payload

A5 supplies the standing-failure obstruction. A representation-bearing gauge realization is admissible only if admitted anomaly-obstruction probes have quotient-trivial obstruction. A nonzero obstruction means gauge-related local descriptions assemble into different boundary-detectable standing ledgers. This is not physical gauge structure; it is failure of gauge standing.

SM-2 uses A5 to classify anomaly equations and anomaly-cancellation calculations as downstream displays of an obstruction-vanishing role. The false primitive deleted is anomaly algebra as merely optional bookkeeping or as primitive structure by formula alone. The carrier is the anomaly-obstruction ledger.

A12.7 A6: gauge-vacuum payload

A6 supplies the bridge to weak-boson standing. Gauge-vacuum compatibility requires that the non-null vacuum profile preserve local transition organization, standing-preserving transport comparison, loop-residue structure, representation-response witnesses, charge-witness ledgers, particle persistence, field-carrier mediation, quotient comparison, bounded transport, boundary trace, deformation-response baseline, and restoration baseline. A vacuum candidate that destroys or hides these structures is not a compatible physical vacuum for that gauge realization.

SM-2 uses A6 to distinguish symmetry-breaking description from primitive gauge destruction. Vacuum-mediated response reduction may later be specialized by Higgs-like mechanisms, condensates, order parameters, vev-like coordinates, residual gauge structures, or mass-generation profiles. Those specializations are licensed routes, not primitive gauge identities.

A12.8 SR1: simultaneous constraint collapse

SR1 proves that the admissible gauge arena cannot remain freely factorized once product, representation, charge, anomaly, vacuum, residual, transport, and deformation ledgers must describe one standing-bearing realized physical interior. SM-2 uses this result to block free recombination of gauge-sector pieces. A product-sector ledger, representation ledger, charge ledger, anomaly ledger, or residual-sector normal form can be locally admissible while failing as a simultaneous realization of all ledgers.

The false primitive deleted is factor-by-factor gauge construction as if each ledger could be promoted independently. The carrier consequence is simultaneous ledger coupling.

A12.9 SR2: product-sector compression

SR2 applies simultaneous collapse to product-sector organization. Raw factor count is not product role-rank. Product-sector organizations survive only when their standing-bearing roles remain quotient-distinct, representation-compatible, charge-compatible, anomaly-neutral, vacuum-supported, residual-stable, and deformation-persistent. SM-2 uses SR2 when treating extra gauge factors and unification claims: a new product label is not automatically a primitive product carrier.

A12.10 SR3: representation-response compression

SR3 compresses representation response over the SR2 product survivor class. Algebraic representation language is licensed only as a downstream refinement of representation-response witnesses. Arbitrary representation labels, dimensions, bases, tensor products, conjugates, response tags, and multiplicities do not carry primitive physical structure. SM-2 uses SR3 for representation-dependent gauge skins and for the handoff to matter standing in SM-3.

A12.11 SR4: Abelian charge-role compression

SR4 compresses Abelian charge-coordinate standing. Raw charge numbers, normalizations, weights, character notation, discrete-coordinate notation, and periodic-coordinate labels do not carry primitive charge standing. SM-2 uses SR4 in photon standing and in dark photon / extra $U(1)$ branch certificates. A new charge-coordinate assignment is not a primitive carrier unless the charge-witness ledger closes.

A12.12 SR5: anomaly-ledger compression

SR5 compresses anomaly-neutrality ledgers over the SR4 charge survivor class. Raw trace sums, charge-power equations, mixed terms, congruence rules, determinant phases, currents, measure factors, counterterms, inflow terms, and regularization choices are downstream coordinates on standing-obstruction neutrality. SM-2 uses SR5 for gauge anomaly branch certificates and for extra gauge factors.

A12.13 SR6: residual/vacuum compression

SR6 compresses residual/vacuum-sector normal forms. It does not license raw residual subgroup notation, gauge-fixing expressions, vacuum representatives, order-parameter coordinates, vev-like coordinates, scalar offsets, residual charge labels, residual representation labels, or residual anomaly equations as primitive residual structure. SM-2 uses this result for weak-boson standing and for the electroweak split boundary.

A12.14 SR7: Standard Model realization rigidity

SR7 assembles product, representation, charge, anomaly, and residual survivor ledgers into the capstone realization-rigidity theorem. SM-2 uses SR7 to receive G_{SM} as recovered gauge-product presentation

up to admissible redescription. It does not use SR7 to derive numerical gauge couplings or the full empirical parameter table.

A12.15 Gauge ontology and coupling-status support

The gauge ontology support blocks an ontic reading of gauge symmetry. Gauge symmetry organizes admissible descriptions; it does not add primitive physical stuff. The coupling-status support blocks the promotion of raw gauge-coupling values into primitive gauge standing. Gauge couplings may be structurally important transport/calibration coordinates, and their invariant overlap constraints may be meaningful, but numeric values do not self-ground the gauge carrier.

A13 Proof-Sketch Ledger for Main SM-2 Theorems

A13.1 Gauge Skin Non-Promotion proof sketch

The proof has three stages. First, SM-1 supplies the general skin-identification and no-promotion form. Second, SM-2 specializes the skin tests to gauge dependence, representation-chart dependence, scheme dependence, measurement-route dependence, detector-route dependence, perturbative-route dependence, projection-interface dependence, and lack of role invariance. Third, the carrier-certificate rule says that any object satisfying those tests can be promoted only by an independent gauge carrier certificate. Therefore a gauge skin without such a certificate cannot be primitive.

A13.2 Boundary-Trace Gauge Role proof sketch

The proof receives R8. If local physical presentations must preserve field-carrier mediation, particle persistence, quotient comparison, bounded transport, and boundary witnesses, then arbitrary local presentation freedom cannot carry standing. Boundary trace fixes the comparison requirement. The resulting role is gauge-order standing. The proof is imported, but SM-2 uses it as the first local carrier stage.

A13.3 Gauge Transition Carrier proof sketch

The proof receives A1. Once boundary-trace gauge-order standing exists, transitions among admitted local presentations must have identity, inverse, restriction, partial composition, associativity, and overlap coherence. Without these, presentation freedom either becomes arbitrary label exchange or destroys standing under composition. The transition groupoid is therefore the minimal carrier for local gauge redescription.

A13.4 Connection Transport Carrier proof sketch

The proof receives A2. Transition maps alone do not compare carrier content along continuation. A standing-preserving comparison assignment is required. If comparison depends on raw representatives, global gauge fixing, or arbitrary path choice, a hidden selector is imported or quotient stability fails. Thus the connection role is the minimal transport carrier. Gauge potentials are licensed as representatives of this role.

A13.5 Curvature/Holonomy Carrier proof sketch

The proof receives A3. Once transport comparison exists, loop comparison can be trivial or nontrivial. A nontrivial residue is admissible only if it is boundary-detectable, quotient-stable, same-domain, transport-witnessed, and role-preserving. This yields curvature/holonomy obstruction standing. Field-strength notation displays the role; it does not create it.

A13.6 Representation/Charge Carrier proof sketch

The proof receives A4 plus SR3 and SR4. Gauge transport must act on field and particle content without destroying persistence identity or quotient comparison. This forces representation-response standing. Charge enters as a boundary-detectable witness of that response. Raw labels and normalizations collapse unless the witness ledger distinguishes them.

A13.7 Anomaly Obstruction Carrier proof sketch

The proof receives A5 plus SR5. Representation-bearing gauge structures must assemble consistently under local-to-global continuation. A nontrivial anomaly obstruction means that gauge-related descriptions produce different boundary-detectable standing ledgers. That is standing failure. Anomaly cancellation is therefore an admissibility requirement for same-scope survivor standing.

A13.8 Gauge-Vacuum Compatibility proof sketch

The proof uses A6 plus SR6. A gauge structure cannot be supported by a vacuum candidate that destroys transition organization, transport comparison, loop-residue standing, representation response, charge witnesses, anomaly neutrality, particle persistence, field-carrier mediation, quotient comparison, bounded transport, boundary trace, deformation baseline, or restoration baseline. Gauge-vacuum compatibility is therefore a carrier condition for weak response and residual-sector standing.

A13.9 Photon carrier proof sketch

Photon standing is obtained by specializing gauge-order, transport, curvature, representation/charge response, and gauge-vacuum compatibility to the unbroken electromagnetic response route. The photon field, polarization, detector event, and Coulomb display are licensed representatives or witnesses. Primitive photon standing, when certified, is charge-response transport standing.

A13.10 Gluon carrier proof sketch

Gluon standing is obtained by specializing the non-Abelian color-transport side of the recovered gauge product. The perturbative gluon line and color-flow diagram are useful route displays; the jet is a detector projection; none supplies primitive isolated-gluon standing. Primitive gluon standing descends to color-transport carrier discipline.

A13.11 Weak-boson carrier proof sketch

Weak-boson standing requires weak-chiral response, gauge-vacuum compatibility, and Higgs bridge/-fixation discipline. The pole mass and width witness downstream mass-response and route behavior. The vev and weak mixing angle are not primitive weak-boson grounds. Primitive weak-boson standing descends through weak-chiral/gauge-vacuum/Higgs-bridge carrier discipline.

A14 Full Classification Certificate Dossiers

Each dossier follows the same format: submitted claim, fixed target, same-scope test, skin test, carrier candidate, branch verdict, and falsifier.

A14.1 Photon dossier

Submitted claim: the photon is primitively the electromagnetic field quantum. The fixed target is D_{gauge} with charge-response and gauge-vacuum ledgers preserved. The field-quantum label passes the skin test because it depends on representative field language and measurement route. The carrier candidate is C_γ , unbroken charge-response transport. The branch verdict is **Carrier** for the certified carrier and **Rep**, **Meas**, or **Proj** for its displays. The falsifier is a same-scope primitive photon carrier outside charge-response transport.

A14.2 Gauge potential dossier

Submitted claim: A_μ is the primitive gauge object. The fixed target is the same gauge arena. The object passes the skin test because it is gauge-presentation dependent and connection-chart dependent. The carrier candidate is the connection transport role, not the potential symbol. The verdict is **Rep** or **Trans**; primitive status fails absent independent certificate.

A14.3 Field-strength dossier

Submitted claim: $F_{\mu\nu}$ is primitive gauge standing. The item passes the skin test because it is a notation for curvature/holonomy residue. The carrier candidate is $\Omega_{\partial D}$. The verdict is licensed representative of obstruction standing. A falsifier must supply same-scope obstruction standing outside the transport-holonomy role.

A14.4 Polarization dossier

Submitted claim: a polarization state is primitive photon identity. The item passes the skin test because it is representative and measurement-route dependent. The carrier remains charge-response transport. The verdict is representative or measurement skin. A same-scope primitive polarization carrier would require redescription-invariant role closure independent of carrier and measurement route.

A14.5 Gluon dossier

Submitted claim: a gluon is a primitive free particle like a photon. The same-scope test preserves the color-transport gauge arena. The free-particle assertion is not automatically target update; it is a

burden if it asserts same-scope primitive isolated standing. The skin test classifies perturbative lines, color labels, jets, and color-flow diagrams as route or projection displays. The carrier candidate is $C_{\text{color-transport}}$. The falsifier is a certified same-scope free-gluon primitive outside color transport.

A14.6 Jet dossier

Submitted claim: a jet is primitive gluon standing. The jet passes the skin test because it is detector-route and projection dependent. It may witness color transport and hadronization routes. It does not ground isolated gluon identity. The verdict is **Proj** or **Route**.

A14.7 W boson dossier

Submitted claim: the W boson is primitively its pole mass and charged current label. The pole mass is a measurement witness and mass-response skin. The charged-current label is a weak response representative. The carrier is weak-chiral response under gauge-vacuum/Higgs bridge discipline. The falsifier is a primitive W carrier outside that package while preserving D_{gauge} .

A14.8 Z boson dossier

Submitted claim: the Z boson is primitively the neutral weak field after mixing. The field label and mixing coordinate pass the skin test. The carrier is weak neutral response under the same weak-chiral/gauge-vacuum/Higgs bridge package. The verdict is carrier for the certified role and licensed representative or transport/mixing status for the field and angle.

A14.9 Weak mixing angle dossier

Submitted claim: the weak mixing angle is primitive electroweak identity. The angle is basis/transport/parameter dependent and belongs to mixing/transport status. It can witness the electroweak split but cannot create primitive weak-boson standing. The falsifier would be a proof that a specific angle coordinate is invariantly required as primitive transition identity independent of carrier and measurement route.

A14.10 Pole mass dossier

Submitted claim: a gauge-boson pole mass grounds primitive identity. Pole mass is a measurement witness of mass-response standing. For W/Z it descends through $H_{\text{br/fix}}$ -licensed mass-response; for photon/gluon claims it belongs to the relevant structural route or burden. The verdict is **Meas**, not primitive standing.

A14.11 Extra vector boson dossier

Submitted claim: a new vector boson supplies a new primitive gauge carrier. If only a label or field is supplied, primitive promotion fails. If a resonance or event excess is supplied, the status is measurement/route or projection/burden. If a new carrier inventory is introduced, target update is available. If same scope is preserved and a carrier certificate closes outside C_{gauge} , local closure fails.

A14.12 Dark photon dossier

Submitted claim: a dark photon is a primitive second electromagnetic carrier. The same-scope test asks whether charge-response standing, anomaly/vacuum ledgers, and target arena are preserved. Kinetic mixing is transport or projection status unless independently certified. A hidden sector usually pressures target-update classification. The falsifier is an independent same-scope electromagnetic primitive outside charge-response standing.

A14.13 Extra $U(1)$ dossier

Submitted claim: an extra $U(1)$ factor is a same-scope primitive gauge factor. The certificate must preserve charge lattice, representation response, anomaly neutrality, vacuum compatibility, normalization discipline, and no-hidden-load conditions. Without that, the verdict is target update, burden, failed primitive, or licensed extension.

A14.14 Grand unification dossier

Submitted claim: a unification group is the primitive gauge ground. If the proposal changes the target arena, it is a target update. If it is a unification display or RG route, it is transport or projection status. If it supplies a same-scope carrier certificate preserving all ledgers, it becomes a serious burden or falsifier. An RG crossing alone is never primitive standing.

A14.15 Gauge EFT anomaly dossier

Submitted claim: a gauge-sector EFT coefficient is primitive new gauge physics. EFT coefficients are projection or transport objects unless they resist all projection, route, measurement, and transport classifications. A repeated anomaly may escalate burden status. It becomes a falsifier only with a same-scope primitive carrier certificate outside C_{gauge} .

A15 Standard Physics Orientation

The paper uses standard physics language as an orientation layer. It does not ask the reader to abandon gauge fields, bundle language, curvature, representations, Feynman rules, polarizations, jets, weak currents, EFT coefficients, or BSM model building. It asks a narrower status question: what carries primitive standing under admissible redescription?

A15.1 Gauge redundancy

Standard gauge theory already treats many presentation choices as redundant. SM-2 sharpens that familiar fact into a carrier/skin distinction. Gauge transformations and gauge choices are not denied; they are classified as representatives of gauge-order and transition standing. A gauge choice is useful because it is licensed, not because it is primitive.

A15.2 Curvature and field strength

Field-strength notation often feels more physical than potential notation because it tracks invariant response. SM-2 agrees that curvature/holonomy has a stronger standing role than a raw potential. The primitive role is still the obstruction carrier, not the formula by itself. This is why field-strength notation is licensed but not self-grounding.

A15.3 Perturbative quanta and detector routes

Perturbative particles are indispensable in calculations. SM-2 does not demote their calculational value. It separates perturbative route status from primitive standing. This matters most for gluons, where perturbative labels, parton-level diagrams, and jets are legitimate but do not by themselves establish primitive isolated-particle standing.

A15.4 Electroweak phenomenology

The W and Z are among the best-confirmed structures in modern physics. SM-2 preserves that phenomenology. Its claim is that pole masses, widths, decay channels, detector events, and weak mixing coordinates are witnesses and routes of weak-chiral/gauge-vacuum/Higgs-bridge standing, not primitive identity grounds.

A15.5 EFT and BSM practice

EFT and BSM model building remain powerful discovery tools. SM-2 classifies their outputs by primitive-status role. A coefficient, resonance, extra vector label, hidden-sector field, or unification display can be a route, projection, extension, target update, burden, or falsifier. The classifier is not anti-BSM; it is anti-unlicensed primitive promotion.

A16 Gauge Branch Algorithm

For operational use, a submitted gauge claim is evaluated by the following branch sequence.

1. Fix D_{gauge} and state the role bundle allegedly preserved.
2. Test for target update: changed gauge product, representation ledger, charge lattice, anomaly ledger, vacuum regime, Higgs boundary, source state, comparison arena, or hidden-sector inventory.
3. If no target update is present, test for carrier certificate.
4. If the certificate closes, assign **Carrier**.
5. If the claim descends to field notation, gauge choice, transport, curvature display, representation label, charge witness, measurement, route, projection, coupling coordinate, or EFT coefficient, assign the registered derivative status.
6. If primitive standing is asserted but the certificate does not close, assign **Burden**.
7. If primitive standing is asserted from label, notation, measurement, route, or projection alone, assign **FailPrim**.
8. Record the prediction-register item and falsifier.

A17 Gauge Sector Release Certificate

The SM-2 package released to SM-6 consists of:

1. C_{gauge} , the local gauge carrier register;
2. the gauge skin atlas;
3. the photon, gluon, and weak-boson carrier certificates;
4. the gauge BSM/EFT branch certificates;
5. the gauge prediction register G0–G7;
6. the hostile gauge reconstruction y_{gauge}^* ;
7. the coupling/RG handoff to SM-4;
8. the color-transport handoff to SM-3;
9. the weak-response handoff to SM-5 and SM-6.

A18 Module-by-Module Proof Certificates

This appendix gives the audit certificates that connect the source modules to the SM-2 closure theorem. The certificates are deliberately repetitive: each records input, carrier output, deleted false primitive, nonclaim boundary, and handoff use.

A18.1 R8 certificate

Field	Entry
Input	Realized physical interior with local presentation, boundary trace, and standing-preserving redescription discipline.
Output	Boundary-trace-fixed gauge role.
Carrier introduced	C_{gord} at the role level.
False primitive deleted	Gauge symmetry or gauge presentation as primitive ontology.
Licensed skins	Gauge-related local descriptions and presentation-equivalence language.
Nonclaim boundary	No specific Lie group, no connection, no curvature, no gauge boson, no numerical coupling.
SM-2 use	Provides the first gauge carrier role that A1 turns into transition organization.
Falsifier	A same-scope local presentation system whose gauge standing is boundary-compatible but not gauge-order carrier standing.

A18.2 A1 certificate

Field	Entry
Input	Boundary-trace-fixed gauge role from R8.
Output	Boundary-fixed transition groupoid $\mathbf{Gpd}_\partial(D)$.
Carrier introduced	Compositional transition carrier for local gauge presentations.
False primitive deleted	Local gauge patch, transition notation, or chart multiplicity as primitive standing.
Licensed skins	Local gauges, transition functions, restriction maps, patch language.
Nonclaim boundary	No connection, curvature, representation, anomaly, vacuum, or specific SM gauge product.
SM-2 use	Establishes the compositional base of C_{gord} .
Falsifier	Same-scope gauge-role standing requiring non-groupoid local transition organization while preserving boundary trace.

A18.3 A2 certificate

Field	Entry
Input	$\mathbf{Gpd}_\partial(D)$ and admissible path comparison.
Output	$\nabla_{\partial D}$, the standing-preserving transport-comparison carrier.
Carrier introduced	Connection-like transport carrier.
False primitive deleted	Gauge potential, connection coefficients, or covariant-derivative notation as primitive object.
Licensed skins	A_μ , connection charts, covariant-derivative notation, transport coefficients.
Nonclaim boundary	No numerical coupling values or primitive potential ontology.
SM-2 use	Grounds the status of gauge potentials and propagator setup as transport representatives.
Falsifier	Same-scope primitive gauge-potential standing that preserves all transport comparison without descending to $\nabla_{\partial D}$.

A18.4 A3 certificate

Field	Entry
Input	Connection-like transport comparison and admissible loop comparison.
Output	$\Omega_{\partial D}$, the curvature/holonomy obstruction carrier.
Carrier introduced	Boundary-detectable loop-residue carrier.
False primitive deleted	Field-strength formula or holonomy display as primitive by notation alone.
Licensed skins	$F_{\mu\nu}$, curvature components, loop displays, Wilson-loop-like notation.
Nonclaim boundary	No claim that every symbolic field strength has primitive standing.

SM-2 use	Grounds field-strength status and obstruction language.
Falsifier	Same-scope loop-residue standing outside transport/holonomy obstruction.

A18.5 A4 certificate

Field	Entry
Input	Transition, transport, and obstruction carriers.
Output	Representation-response and charge-witness carrier roles.
Carrier introduced	$\text{Rep}_{\partial D}$ and $\text{ChWit}_{\partial D}$.
False primitive deleted	Representation label, charge number, current label, or detector readout as primitive standing.
Licensed skins	Representation notation, charge values, current expressions, charge-detection routes.
Nonclaim boundary	No derivation of every numerical charge or coupling value at this layer.
SM-2 use	Grounds photon charge-response and prepares product/charge rigidity.
Falsifier	Same-scope charge or representation standing outside response/witness carrier discipline.

A18.6 A5 certificate

Field	Entry
Input	Representation-response and charge-witness carrier roles.
Output	Anomaly-obstruction ledger.
Carrier introduced	Standing-failure obstruction discipline.
False primitive deleted	Raw anomaly equation, trace sum, regularization choice, or repair term as primitive structure.
Licensed skins	Anomaly equations, trace calculations, algebraic cancellation displays.
Nonclaim boundary	No prohibition on target-updating UV completion.
SM-2 use	Tests extra gauge factors, gauge anomalies, and charge/representation survivor claims.
Falsifier	Same-scope anomalous gauge standing preserving all ledgers with a closed certificate.

A18.7 A6 certificate

Field	Entry
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Input	Anomaly-neutral gauge/representation/charge survivor candidate.
Output	Gauge-vacuum compatibility carrier GaugeVac_D .
Carrier introduced	Vacuum-compatible gauge standing.
False primitive deleted	Vacuum coordinate, branch label, broken-phase notation, or vev display as primitive gauge identity.
Licensed skins	Broken/unbroken phase descriptions, vacuum route language, residual-sector displays.
Nonclaim boundary	No numerical Higgs-sector or electroweak parameter closure.
SM-2 use	Supplies weak-boson carrier boundary and Higgs bridge interface.
Falsifier	Same-scope weak gauge standing outside gauge-vacuum compatibility.

A19 SR Rigidity Crosswalk

SR	Survivor projection	Primitive promotion blocked	SM-2 consequence
SR1	shared-ledger simultaneous survivor	freely factorized product/representation/charge/anomaly/vacuum variations	gauge claims cannot vary one ledger while pretending to preserve the whole target.
SR2	product-sector survivor	group-factor labels as primitive	extra gauge factors require carrier certificates or target updates.
SR3	representation-response survivor	representation notation as primitive	matter/gauge response labels descend to representation carrier.
SR4	Abelian charge survivor	raw charge coordinate or normalization as primitive	photon charge-response is carrier-indexed.
SR5	anomaly-ledger survivor	anomaly equations or regularization displays as primitive	gauge anomalies are obstruction/burden/update, not automatic primitives.
SR6	residual/vacuum survivor	vacuum branch or residual coordinate as primitive	W/Z standing requires gauge-vacuum and Higgs bridge discipline.
SR7	joint SM-like survivor	independently selected gauge skeleton	SM-2 receives recovered G_{SM} as structural survivor, not primitive group postulate.

A20 False Primitive Deletion Ledger

False primitive	Deletion reason	Licensed replacement	Serious falsifier
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Gauge symmetry ontology	gauge-related descriptions are representationally redundant	boundary-trace gauge-order carrier	non-redundant ontic gauge structure preserving same observable quotient.
Gauge potential	local/gauge-dependent representative	connection-like transport carrier	invariant primitive potential outside transport.
Gauge fixing	selected description/calculation route	gauge-fixing skin	gauge choice with independent primitive standing.
Ghost field	calculational artifact of gauge-fixing route	route/calculation skin	physical primitive ghost carrier under same gauge target.
Propagator	gauge, scheme, or perturbation dependent	calculation route	primitive propagator identity independent of route.
Polarization	basis/measurement representative	representative/witness skin	primitive polarization standing independent of carrier.
Field strength formula	representative of obstruction	curvature/holonomy carrier	primitive formula standing outside obstruction role.
Group factor label	product-sector notation	product survivor role	same-scope new primitive factor preserving all ledgers.
Coupling value	calibration/scale/scheme dependent	transport ledger	primitive gauge carrier by numerical value alone.
RG crossing	projection or transport route	transport or projection display	crossing that supplies independent carrier standing.
Jet	detector/hadronization route	color-transport evidence route	isolated primitive gluon certificate.
W/Z pole mass	measurement and mass-response display	weak carrier witness	primitive weak identity by pole value alone.
EFT coefficient	basis, matching, or projection coordinate	projection or transport status	coefficient requiring primitive carrier under fixed target.

A21 Future-Claim Audit Scenarios

A21.1 Scenario 1: precision photon anomaly

A precision photon-sector anomaly first enters as measurement witness or route stress. If it depends on detector calibration, low-energy projection, or transport scheme, it remains derivative. It becomes burden only if those statuses fail. It becomes target update if it changes the comparison arena. It becomes falsifier only if it supplies a same-scope primitive photon carrier outside charge-response standing.

A21.2 Scenario 2: claimed free gluon observation

A claimed free gluon is a high-severity gauge-sector event. SM-2 does not reject it by slogan. It asks whether the claim is a jet/projection route, a detector misclassification, a target update, or an actual same-scope primitive color carrier outside C_{coltrans} . Only the last branch falsifies local color-transport closure.

A21.3 Scenario 3: W mass deviation

A W mass deviation is first a measurement witness, route stress, or mass-response/projection issue. It may become burden if it coheres across routes. It becomes target update if it changes gauge-vacuum/Higgs-bridge target conditions. It is not primitive W standing by pole value alone.

A21.4 Scenario 4: new narrow vector resonance

A new vector resonance enters the extra vector branch certificate. It may be detector route, EFT projection, extension disposition, target update, burden, failure, or falsifier. SM-2 does not decide by name; it decides by carrier certificate and target preservation.

A21.5 Scenario 5: extra $U(1)$ model

A written $U(1)'$ factor is a model surface. Same-scope survival requires preservation of representation, charge, anomaly, vacuum, and boundary ledgers. Otherwise it is target update or extension. Primitive standing by group label alone fails.

A21.6 Scenario 6: unification crossing

A coupling crossing is a transport or projection display unless an independent carrier certificate is supplied. If the unification proposal changes the target arena, it is target update. If it claims same-scope primitive ground by a plot alone, it fails primitive promotion.

A21.7 Scenario 7: gauge EFT coefficient

An EFT coefficient is operator-basis and matching dependent unless the submitted claim proves otherwise. It can raise burden status if repeated anomalies cohere. It becomes a primitive survivor only if projection, transport, measurement, and target-update classifications are all defeated and a carrier certificate closes.

A21.8 Scenario 8: hidden vector sector

A hidden vector sector normally changes carrier inventory or source state and is therefore target update or extension. It does not refute same-scope closure unless it preserves D_{gauge} and supplies an independent same-scope gauge carrier outside C_{gauge} .

A22 Glossary of Gauge-Sector Terms

Term	SM-2 meaning
Gauge-order carrier	standing-bearing role for boundary-fixed local presentation equivalence.
Transition groupoid	compositional organization of local/restriction gauge transitions.
Transport carrier	standing-preserving comparison along admissible routes.
Curvature/holonomy carrier	boundary-detectable loop-residue obstruction.
Representation response	admissible response of physical carriers under gauge transport.
Charge witness	boundary-detectable charge-response standing attached to representation response.
Anomaly ledger	fail-closed obstruction discipline for gauge/representation/charge standing.
Gauge-vacuum compatibility	residual and vacuum-support discipline for gauge standing.
Photon carrier	unbroken electromagnetic charge-response transport.
Gluon carrier	non-asymptotic non-Abelian color-transport standing.
Weak-boson carrier	weak-chiral response licensed by gauge-vacuum and Higgs bridge/fixation discipline.
Gauge skin	gauge-sector object whose standing depends on presentation, gauge, basis, scheme, route, projection, or model label.
Target update	changed gauge carrier inventory, invariant bundle, boundary, source state, arena, or regime.
Burden	live same-scope primitive assertion without closed certificate.

A23 Full Gauge Claim Atlas

The following atlas expands the gauge skin and carrier register into operational claim classes. Each row identifies the submitted surface, the carrier question, the likely derivative status, and the exact escalation path required for primitive standing.

Claim surface	Carrier question	Expected classifier status	Escalation path
Gauge potential	Does it preserve transport role independent of gauge chart?	licensed representative or transport coordinate	transport carrier certificate outside $\nabla_{\partial D}$.

Gauge transformation	Does it carry standing beyond redescription relation?	gauge-order skin	same-scope primitive gauge-order outside transition organization.
Gauge fixing	Does a selected representative carry invariant role?	skin / calculation route	no-hidden-selector proof plus carrier closure.
Ghost field	Is the field an artifact of gauge-fixed quantization or an independent carrier?	calculational skin	physical carrier certificate preserving all ledgers.
Field strength	Is the formula the carrier or a display of obstruction?	curvature representative	obstruction standing outside $\Omega_{\partial D}$.
Wilson-style loop display	Does the display witness loop residue?	projection / route display	closed loop-residue carrier outside holonomy role.
Generator chart	Does generator notation survive chart changes?	notation skin	role closure independent of chart.
Gauge group label	Does label encode recovered product role or new target?	representative, burden, or update	product-role certificate preserving all ledgers.
Photon field quantum	Does field-quantum label ground EM standing?	licensed representative	primitive photon carrier outside charge response.
Photon polarization	Does polarization ground identity?	measurement / representative skin	invariant role independent of route.
Coulomb display	Does low-energy display ground primitive EM carrier?	projection route	primitive EM carrier outside charge response.
Gluon line	Does perturbative line ground asymptotic standing?	calculation route	same-scope free-gluon carrier certificate.
Jet	Does detector topology ground isolated gluon?	detector projection	primitive isolated color carrier.
Color flow	Does diagram ground color transport?	projection display	carrier outside color-transport role.
W field	Does notation ground weak charged response?	licensed representative	primitive weak carrier outside weak-chiral role.
Z field	Does notation ground weak neutral response?	licensed representative	primitive weak carrier outside weak-chiral role.

Weak mixing angle	Does coordinate ground transition identity?	mixing / transport coordinate	basis-independent primitive transition certificate.
Pole mass	Does measured value ground boson identity?	measurement witness	primitive identity by value plus carrier certificate.
Decay width	Does decay route ground primitive standing?	route / measurement witness	route-independent carrier certificate.
Gauge coupling	Does number ground gauge carrier?	transport/calibration ledger	invariant fixing certificate, SM-4.
RG crossing	Does crossing generate primitive standing?	transport or projection display	strongly isolated carrier-fixing theorem.
Anomaly equation	Is algebraic cancellation primitive?	obstruction display	anomaly carrier outside obstruction ledger.
EFT coefficient	Does coefficient ground new carrier?	projection or transport ledger	carrier pressure outside projection and matching.
Extra vector label	Does label add primitive gauge standing?	branch certificate	same-scope carrier certificate outside C_{gauge} .
Hidden sector	Does it preserve D_{gauge} or change target?	update, extension, burden	target-preserving independent carrier proof.

A24 Detailed Gauge BSM Branch Certificates

A24.1 Extra vector boson branch

A submitted extra-vector claim first enters as a claim surface: a field label, resonance, decay topology, EFT coefficient, hidden-sector mediator, or additional gauge factor. SM-2 asks whether the claim preserves D_{gauge} . If it changes carrier inventory, representation ledger, charge lattice, anomaly ledger, vacuum regime, or source arena, it is target update. If it preserves those ledgers but supplies only a field label or resonance name, primitive promotion fails. If it supplies persistent anomaly-resistant, redescription-invariant role standing not descending to C_{gauge} , it becomes a burden and eventually a falsifier when the carrier certificate closes.

A24.2 Dark photon branch

The dark-photon branch is deliberately not resolved by name. The phrase may denote kinetic mixing, a hidden-sector vector, a low-energy projection, a resonance route, or a new carrier inventory. Kinetic mixing is transport or projection status unless it comes with independent charge-response standing. A hidden-sector vector is typically a target update if the carrier inventory changes. A same-scope dark photon falsifier must preserve the Standard Model charge-response carrier while supplying independent primitive electromagnetic standing outside C_{chresp} .

A24.3 Extra $U(1)$ branch

An extra Abelian factor is the sharpest product-sector test. A new $U(1)$ label is not a primitive carrier. The certificate must preserve product-role compression, charge-ratio rigidity, anomaly neutrality, residual/vacuum compatibility, normalization discipline, and no-hidden-load conditions. If any of those ledgers changes, the claim is target update or extension. If all are preserved and the new factor remains quotient-distinct with a closed carrier certificate, SM-2's local product-sector branch is falsified.

A24.4 Grand unification branch

A unification group may function as a target update, a larger-domain structure, a transport/unification skin, or a burden. SM-2 does not adjudicate the numerical or phenomenological viability of grand unification. It asks whether the unification claim is same-scope Standard Model gauge standing. If the unification group changes the target arena, it is not a same-scope survivor. If it is a projection or RG-display claim, it is transport status. If it preserves D_{gauge} while supplying an independent primitive group carrier outside the recovered gauge product, it becomes a falsifier.

A24.5 Gauge EFT anomaly branch

A gauge-sector EFT anomaly is serious evidence only when its claim surface is specified. A coefficient may depend on operator basis, matching convention, renormalization scheme, low-energy projection, or measurement route. Repeated anomaly patterns can escalate burden status, but primitive standing requires more: the anomaly must resist measurement, route, projection, and transport classification under fixed D_{gauge} , and must supply a closed carrier certificate outside C_{gauge} . SM-2 therefore treats anomalies as discovery-positive but primitive-status disciplined.

A24.6 Massive hidden vector branch

A massive hidden vector may be a detector route, a resonance surface, a kinetic-mixing projection, a hidden-sector target update, or a burden. Its mass does not ground primitive gauge standing. Its field label does not ground primitive gauge standing. Its hidden-sector interpretation usually changes the target. A same-scope primitive hidden vector would need to preserve gauge product, charge, representation, anomaly, vacuum, and Higgs/fixation ledgers while remaining independent of the SM gauge carrier network.

A24.7 Composite gauge scenario branch

A composite gauge scenario is not rejected by name. It is classified by target relation. If compositeness reconstructs a carrier while preserving D_{gauge} , it must prove that the reconstructed carrier displaces or supplements C_{gauge} without changing the target. If compositeness changes the admissibility boundary or carrier inventory, the result is a target update. If it is a model surface or EFT route, it is extension or projection status.

A25 Gauge-Sector Dependency and Audit-Sensitivity Map

Theorem area	Closure status	Audit sensitivity	Reason
Gauge-order carrier	closed	low	supported by R8 and A1.
Transport carrier	closed	low	supported by A2.
Curvature/holonomy carrier	closed	low/medium	formulas are salient, but the carrier is the obstruction role.
Charge-response / photon	closed	medium	depends on A4, PSM4, SR4, and L1 charge/EW bookkeeping.
Color-transport / gluon	closed	high dependency sensitivity	depends on the Yang–Mills endpoint package.
Weak-chiral / W/Z	closed	medium/high dependency sensitivity	depends on A6, PSM6, SR6/SR7, L1 equivalence, and Higgs/mass-fixation support.
Anomaly ledger	closed	medium	depends on A5, PSM5, SR5, and L1 anomaly/Yukawa support.
Gauge-vacuum carrier	closed	medium	depends on A6, PSM6, SR6, and Higgs/mass-fixation support.
Local primitive-status deletion	closed	medium/high dependency sensitivity	follows from the complete carrier-certificate ledger.
Gauge BSM branch certificates	branch logic closed	medium/high dependency sensitivity	named models require separate instantiation.
Global SM endpoint	not claimed in SM-2	not applicable	belongs to SM-6.

Audit sensitivity does not mark an open theorem. It marks how much proof load is carried by imported support modules that a referee may wish to inspect.

A26 Complete Component Audit Forms

Each form below uses the SM-1 audit grammar. The fields are: submitted object, fixed target, carrier candidate, skin tests, descent route, target-update boundary, burden condition, and terminal status.

A26.1 Gauge potential audit form

Field	Entry
Submitted object	Gauge potential notation, locally written as A_μ or an equivalent connection representative.
Fixed target	D_{gauge} , with boundary-trace transition organization and standing-preserving transport comparison fixed.
Carrier candidate	$\nabla_{\partial D}$, not the local potential components.
Skin tests	Gauge choice, chart choice, local trivialization, normalization, and calculation route affect the representative.
Descent route	The potential descends to connection-like transport comparison.
Target-update boundary	A new transport law or changed gauge arena may update the target; a new notation does not.
Burden condition	Primitive potential standing must be invariant under admissible redescription and not reducible to transport comparison.
Terminal status	Licensed representative / transport coordinate.

A26.2 Gauge fixing and ghost audit form

Gauge fixing and ghosts are treated together because both are bound to calculation route. Gauge fixing chooses a representative within a redundant presentation class. Ghost fields arise in gauge-fixed quantization and preserve calculational consistency. Neither is primitive gauge standing under D_{gauge} . A ghost-sector primitive claim would need to show physical carrier standing outside the gauge-fixing route while preserving the same gauge target; otherwise it is a route/calculation skin or target update.

A26.3 Propagator audit form

A propagator is a powerful calculation object. It depends on representation, gauge choice, perturbative expansion, and boundary/route assumptions. Its lawful use is preserved as a route/calculation skin. The primitive carrier is not the propagator form but the gauge carrier package that licenses the field and transport structure whose perturbative behavior the propagator displays.

A26.4 Polarization audit form

Polarization language is retained as representative and measurement language. It is basis-dependent and route-dependent. It can witness photon or weak-boson response in a detector or scattering process. It does not create primitive gauge-boson standing. A primitive polarization claim would need redescription-invariant role closure independent of field/presentation carrier discipline.

A26.5 Pole mass and width audit form

Pole masses and widths are measurement witnesses and route statuses. They are indispensable empirical anchors, but they are not primitive gauge-boson identity grounds. Photon masslessness, W/Z pole values,

and widths receive status only through their carrier package: charge-response or weak-chiral/gauge-vacuum/Higgs-bridge standing. A pole value can pressure the classifier as a measurement anomaly, but the value does not self-promote.

A26.6 Color-flow and jet audit form

Color-flow diagrams and jets are downstream displays of color-transport standing. They are not primitive color objects. A jet is detector-facing and hadronization-route dependent. A color-flow diagram is a bookkeeping/projection display. Their evidentiary force is preserved by SM-2, but their primitive promotion is denied unless a same-scope carrier certificate closes outside C_{coltrans} .

A26.7 Coupling and RG audit form

Coupling values and RG flows are assigned to transport/calibration status. The gauge-coupling source papers block primitive standing by raw value. SM-2 records this only as boundary to SM-4. A coupling anomaly can become evidence of route stress, projection mismatch, or target update; it is not primitive gauge standing by number alone.

A26.8 Anomaly audit form

Anomaly language has two lawful roles. It can be an obstruction ledger inside same-scope gauge standing, or it can signal target-update pressure. Raw anomaly equations, trace sums, regularization choices, and counterterms are not primitive. A gauge anomaly falsifies SM-2 only if it supplies same-scope primitive carrier standing outside C_{gauge} while preserving all fixed ledgers.

A27 Evidence Severity Tiers for Gauge Claims

Tier	Gauge-sector meaning	Effect on SM-2
0	new label, notation, model name, gauge choice, or parameterization	no endpoint pressure; classify as representative, route, projection, or failure.
1	single measurement anomaly in photon, gluon, weak, or vector sector	classify as measurement witness, route stress, projection issue, transport correction, burden candidate, or target-update candidate.
2	repeated coherent anomalies across experiments or routes	raises burden status; may motivate BSM branch certificate or target-update test.
3	carrier-pressure evidence resisting measurement, route, projection, and transport classifications	serious burden; requires explicit carrier-certificate search.
4	same-scope primitive gauge survivor outside C_{gauge} with closed certificate	falsifies the local gauge-sector primitive-status deletion theorem.

A28 Denial Obligations for Gauge Counterexamples

A submitted gauge counterexample has standing only if it discharges the following obligations.

1. **Target preservation.** It must preserve D_{gauge} , not merely use familiar gauge vocabulary.
2. **Carrier-inventory preservation.** It must not add a hidden sector, new invariant bundle, or changed source state while pretending to be same-scope.
3. **Derivative-status exclusion.** It must show that the claim is not a field label, gauge choice, polarization, propagator, coupling value, RG coordinate, detector route, jet, color-flow diagram, EFT coefficient, or model name.
4. **Descent defeat.** It must defeat descent to C_{gord} , $\nabla_{\partial D}$, $\Omega_{\partial D}$, $\text{ChWit}_{\partial D}$, C_{chresp} , C_{coltrans} , C_{weakch} , and GaugeVac_D , as relevant.
5. **Carrier certificate.** It must supply role closure, redescription invariance, licensing proof, descent exclusion, no-hidden-selector proof, and gauge-sector closure.
6. **Burden closure.** It must move beyond live burden status into a closed certificate. Until then it is neither accepted as primitive survivor nor dismissed.

These obligations are intentionally strict because the endpoint claim is strict. They do not block new physics. They identify what sort of new physics would count as same-scope primitive survival.

A29 Dependency and Audit-Sensitivity Summary

Theorem area	Closure status	Audit note
Gauge skin non-promotion	closed	follows from SM-1 skin tests plus gauge local-dependence facts.
Boundary-trace gauge role	closed	depends on R8 and the realized-physics bridge.
Transition/transport ladder	closed	source modules A1–A3 supply the load.
Representation/charge witness	closed	supported by A4, PSM4, SR4, and L1 charge/EW bookkeeping; numerical coupling values remain downstream.
Anomaly/vacuum compatibility	closed	obstruction/vacuum discipline is closed under the declared arena; target-updating UV completions remain possible.
Photon carrier	closed	charge-response route is closed without claiming numerical coupling derivation.
Gluon carrier	closed	dependency-sensitive because it imports the Yang–Mills endpoint package.
Weak-boson carrier	closed	dependency-sensitive because it imports L1 equivalence and Higgs/mass-fixation support.
Gauge BSM branch certificates	branch logic closed	named models require separate instantiation.

Local primitive-status deletion	closed	follows from all sector-local carrier certificates.
Global SM endpoint	not claimed	belongs to SM-6.

A30 SM-2 Handoff Package

Recipient paper	Receives from SM-2	Use
SM-3	color-transport carrier; gluon non-asymptotic standing; jet/color-flow status	matter-sector quark/gluon interface and confinement-compatible quark standing.
SM-4	gauge-coupling non-carrier result; RG/coupling transport status	full charges, couplings, and renormalization transport closure.
SM-5	weak-chiral response and gauge side of transition standing	flavor/mixing/generation coordination.
SM-6	C_{gauge} , gauge prediction register, scoped local primitive-status deletion, gauge BSM certificates	final Standard Model carrier exhaustion synthesis.

A31 Publication-Ready Claim Summary

SM-2’s publication-facing claim can be stated without project shorthand:

Within the fixed Standard Model gauge comparison arena, no residual same-scope primitive gauge-sector attribute survives outside the closed gauge carrier network. Photon language is licensed by charge-response and unbroken electromagnetic transport; gluon language is licensed by non-Abelian color transport; weak-boson language is licensed by weak-chiral, gauge-vacuum, and Higgs-bridge response. Field notation, gauge choice, polarization, propagator behavior, mass value, width, longitudinal/Goldstone description, jet, detector event, coupling value, and model label remain representatives, witnesses, transport ledgers, routes, projections, extensions, target updates, burdens, failures, or genuine falsifiers according to the stated certificate conditions.

This summary is intentionally not an empirical no-new-physics claim. It is a scoped primitive-status closure claim for the gauge sector under fixed D_{gauge} .

References

- [1] Amos Jay Maley, *Primitive Carriers, Derivative Skins, and Classifier-Level Prediction Closure in the Standard Model*, SM-1, July 2026. DOI: 10.5281/zenodo.21213269.

- [2] Amos Jay Maley, *Standard Model Carrier Exhaustion Closure: Primitive Closure, Skin Openness, and Prediction Register*, July 2026.
- [3] Amos Jay Maley, *The Standalone Standard Model Structure from First Principles*, May 2026.
- [4] Amos Jay Maley, *R8: Gauge Structure from Boundary Trace Fixation*, 2026.
- [5] Amos Jay Maley, *The Gauge Transition Groupoid Theorem*, A1, 2026.
- [6] Amos Jay Maley, *Connection as Standing-Preserving Transport Geometry*, A2, 2026.
- [7] Amos Jay Maley, *Curvature as Transport-Holonomy Obstruction*, A3, 2026.
- [8] Amos Jay Maley, *Representation Admissibility and Charge-Witness Rigidity*, A4, 2026.
- [9] Amos Jay Maley, *Anomaly as Standing-Failure Obstruction*, A5, 2026.
- [10] Amos Jay Maley, *Gauge-Vacuum Compatibility*, A6, 2026.
- [11] Amos Jay Maley, *Simultaneous Constraint Collapse of the Admissible Gauge Arena*, SR1, 2026.
- [12] Amos Jay Maley, *Product-Sector Rigidity in the Admissible Gauge Arena*, SR2, 2026.
- [13] Amos Jay Maley, *Representation-Response Rigidity in the Admissible Gauge Arena*, SR3, 2026.
- [14] Amos Jay Maley, *Abelian Charge-Ratio Rigidity in the Admissible Gauge Arena*, SR4, 2026.
- [15] Amos Jay Maley, *Anomaly-Ledger Rigidity in the Admissible Gauge Arena*, SR5, 2026.
- [16] Amos Jay Maley, *Residual Vacuum-Sector Rigidity in the Admissible Gauge Arena*, SR6, 2026.
- [17] Amos Jay Maley, *Standard-Model Realization Rigidity in the Admissible Gauge Arena*, SR7, 2026.
- [18] Amos Jay Maley, *Why Gauge Symmetry Cannot Be Ontic*, 2026.
- [19] Amos Jay Maley, *Gauge Coupling Parameters and Forced Uniqueness*, 2026.
- [20] Amos Jay Maley, *Dimensionless Gauge Couplings as Moduli*, 2026.
- [21] Amos Jay Maley, *Lie-Type Gauge Realization Admissibility*, PSM1, 2026.
- [22] Amos Jay Maley, *Product Gauge-Sector Compatibility*, PSM2, 2026.
- [23] Amos Jay Maley, *Representation Algebra Specialization*, PSM3, 2026.
- [24] Amos Jay Maley, *Abelian Charge-Witness Normalization*, PSM4, 2026.
- [25] Amos Jay Maley, *Anomaly Ledger Algebraization*, PSM5, 2026.
- [26] Amos Jay Maley, *Vacuum-Compatible Residual Sector Normal Form*, PSM6, 2026.
- [27] Amos Jay Maley, *Vacuum-sector mass gap for the local gauge-invariant sharp-local Yang–Mills net*, 2026.
- [28] Amos Jay Maley, *Proof Kernel for the Local Gauge-Invariant Sharp-Local Yang–Mills Vacuum-Sector Mass-Gap Theorem*, 2026.
- [29] Amos Jay Maley, *Companion I: Ultraviolet Gate and Route 1 Mass-Gap Chain*, 2026.

- [30] Amos Jay Maley, *Companion II: Lane A Sharp-Local Construction*, 2026.
- [31] Amos Jay Maley, *Companion III: Reconstruction, Non-Triviality, and Faithful-Wilson Universality*, 2026.
- [32] Amos Jay Maley, *Admissibility and the Well-Definedness of Yang–Mills Endpoint Constructions*, 2026.
- [33] Amos Jay Maley, *Equivalence Completion and Forced Collapse of Charged-Current Degrees of Freedom*, 2026.
- [34] Amos Jay Maley, *Equivalence Completion Between Electroweak and Electromagnetic Coupling Bookkeeping*, 2026.
- [35] Amos Jay Maley, *Equivalence Completion for Longitudinal Gauge Modes and Goldstone Degrees*, 2026.
- [36] Amos Jay Maley, *Anomaly Freedom iff Yukawa Realizability under Minimal Chiral Load*, 2026.
- [37] Amos Jay Maley, *Absence of Tree-Level Flavor-Changing Neutral Currents under Admissible Redescription*, 2026.
- [38] Amos Jay Maley, *Hypercharge Normalization and Electric Charge Quantization under Admissible Redescription*, 2026.
- [39] Amos Jay Maley, *Exhaustion and Closure of the Equivalence Space*, 2026.
- [40] Amos Jay Maley, *Mass Fixation and Inertial Response: Why the Higgs Was Structurally Unavoidable*, 2026.
- [41] M. Peskin and D. Schroeder, *An Introduction to Quantum Field Theory*, Addison-Wesley, 1995.
- [42] S. Weinberg, *The Quantum Theory of Fields*, Cambridge University Press, 1995-2000.
- [43] M. D. Schwartz, *Quantum Field Theory and the Standard Model*, Cambridge University Press, 2014.
- [44] S. Weinberg, *Phenomenological Lagrangians*, Physica A 96, 327-340, 1979.
- [45] H. Georgi, *Effective Field Theory*, Annual Review of Nuclear and Particle Science 43, 209-252, 1993.
- [46] A. J. Maley, *somamaley-ux/AASC-Standard-Model-Carrier-Exhaustion-Generation-Cardinality: v0.1.0*, GitHub repository AASC-Standard-Model-Carrier-Exhaustion-Generation-Cardinality, release v0.1.0, 2026. DOI: 10.5281/zenodo.21219417.